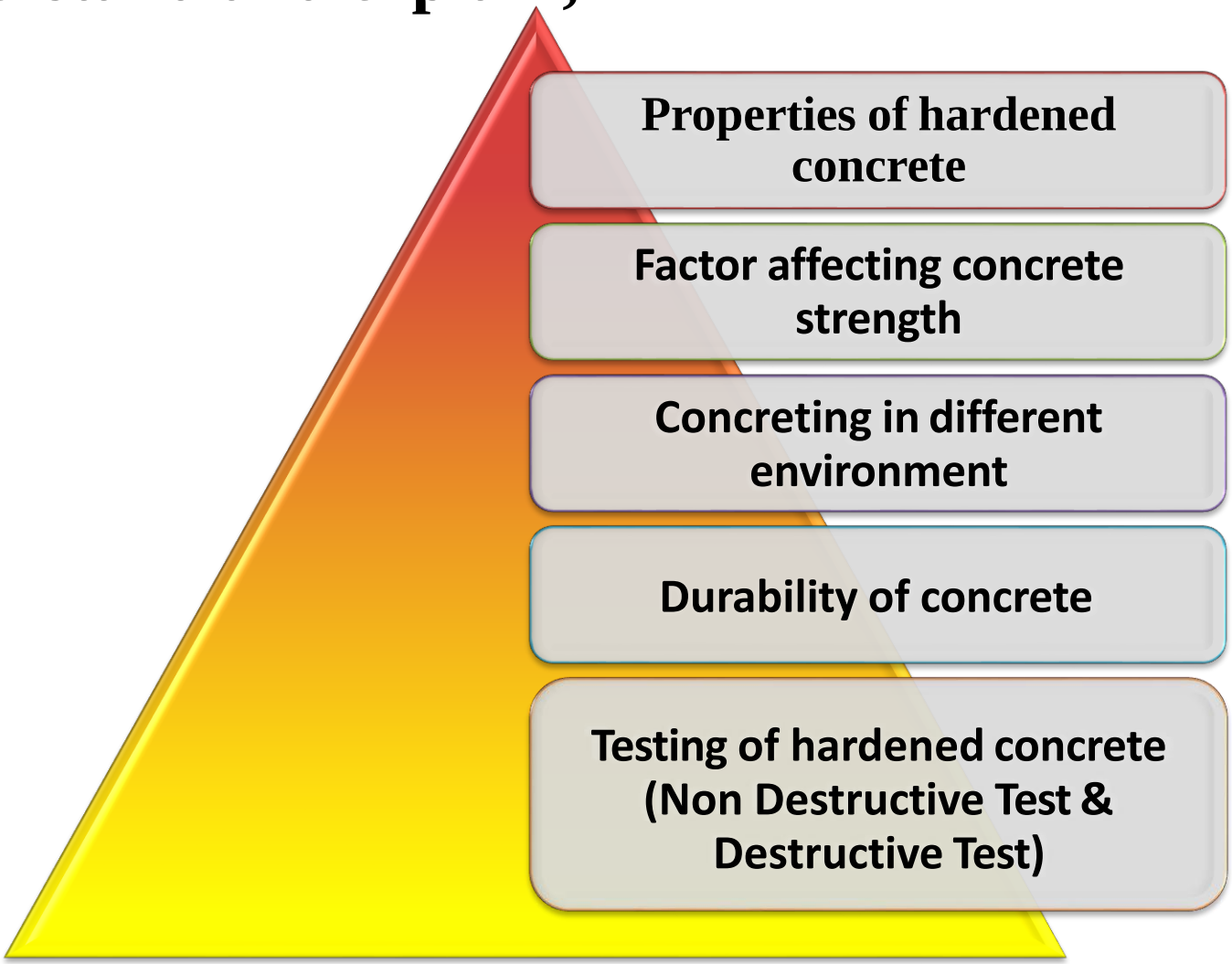
HARDENED CONCRETE





LEARNING OUTCOME

At the end of this lecture, students should be able to understand and explain;



SETTING AND HARDENING

- Dormant Period – time from mixing cement with water and produce plastic workable paste. The paste remain virtually unchanged.
- Initial set – the paste begin to stiffen to such a degree, it becomes unworkable.
- Initial setting time – the time required for the paste to reach the stage.

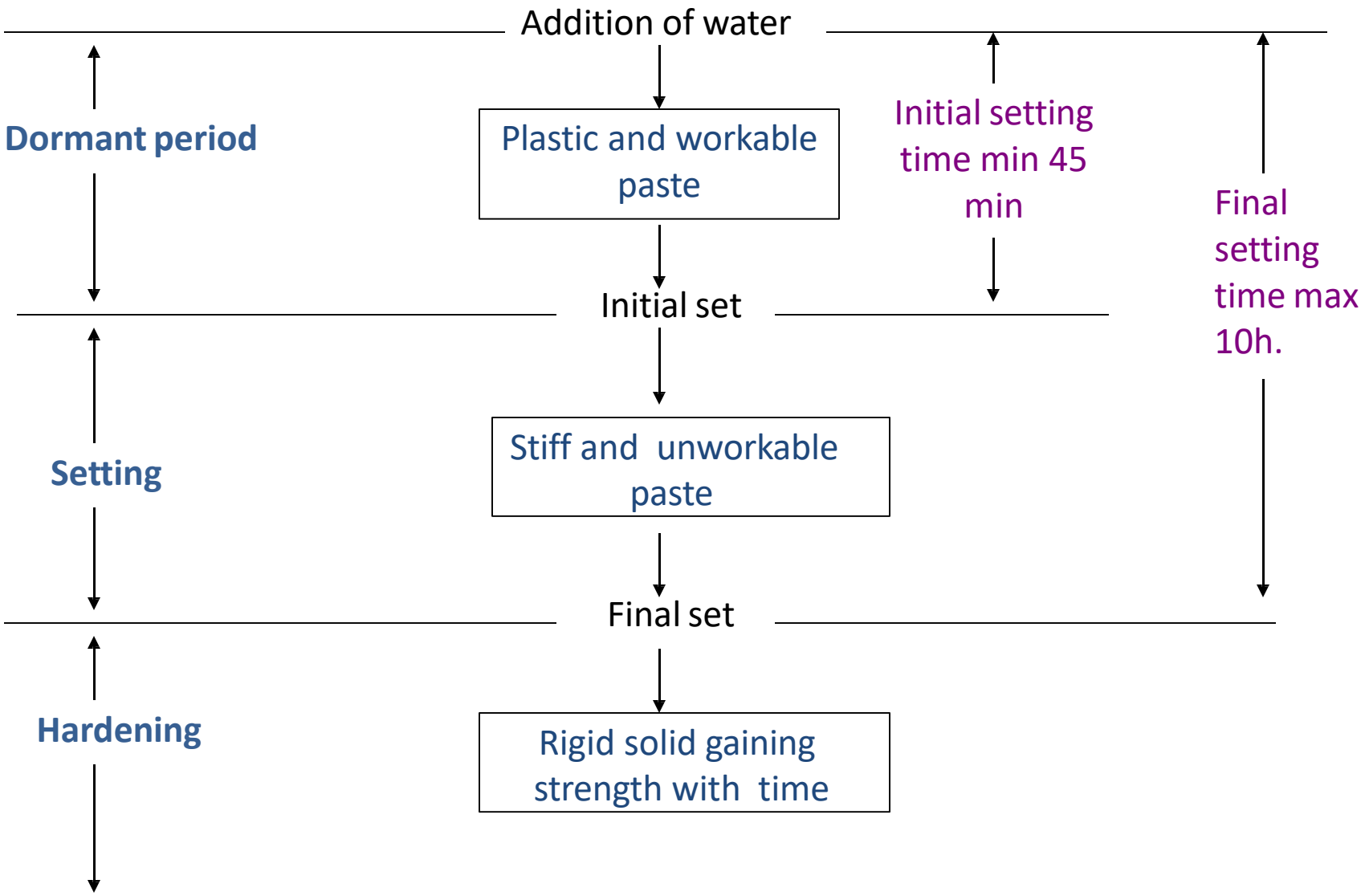
SETTING AND HARDENING

- Final Set – the paste continues to stiffen until a stage is reached when it may be regarded as a rigid solid (complete solidification).

- Final setting time – the time required for the paste to reach the final set stage.

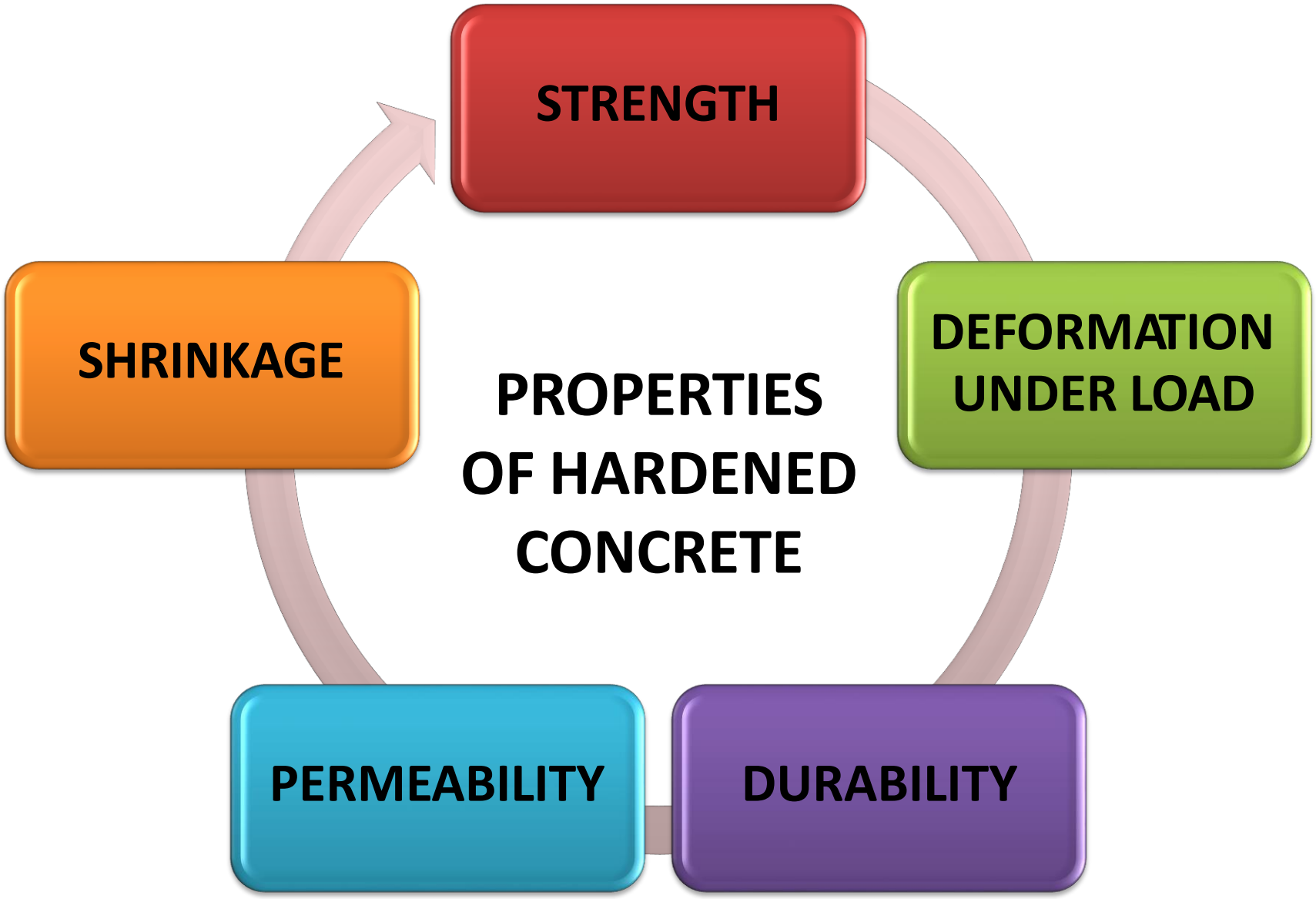
- Hardening – the process with time the hardened paste continues to harden and gain the strength.

Figure 1: Schematic description of setting and hardening of cement paste.



Mechanical Properties

- Properties (instantaneous/time-dependent)
  - Aging (Strength/Stiffness)
  - Creep/Relaxation/Loading Rate
  - Shrinkage
- Strength (uniaxial/multi-axial), Fundamental modes
  - Compression, Tension, Flexure, Shear, Torsion
- Durability
  - Permeability and Pore Structure
  - Freeze-Thaw Resistance
  - Corrosion Resistance
  - ASR, Carbonation, Other Chemical Deterioration


STRENGTH
DEFORMATION UNDER LOAD
DURABILITY
PERMEABILITY
SHRINKAGE
PROPERTIES OF HARDENED CONCRETE

PROPERTIES OF HARDENED CONCRETE

1. STRENGTH

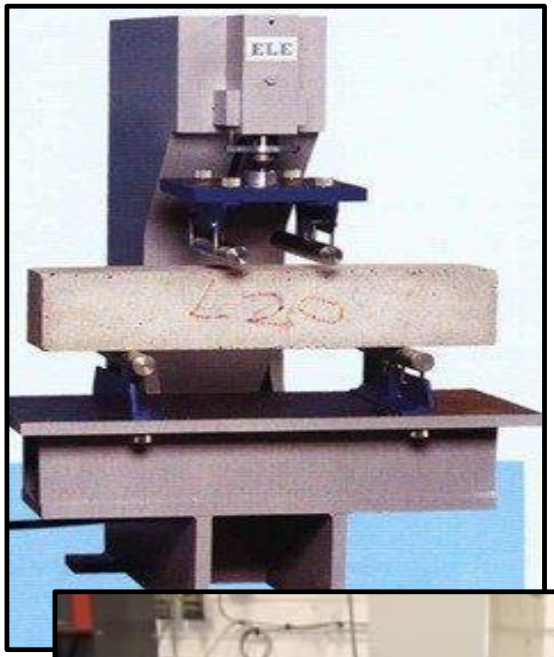
- Strength of concrete is defined as the max stress it can resist or the max it can carry.
- Cubes, cylinders and prisms are the 3 type of compression test specimens.
- Flexural tensile test is used to estimate the load at which the concrete members may crack.
- Compressive Strength taken as the maximum compression load it can carry per unit area.

COMPRESSION TEST

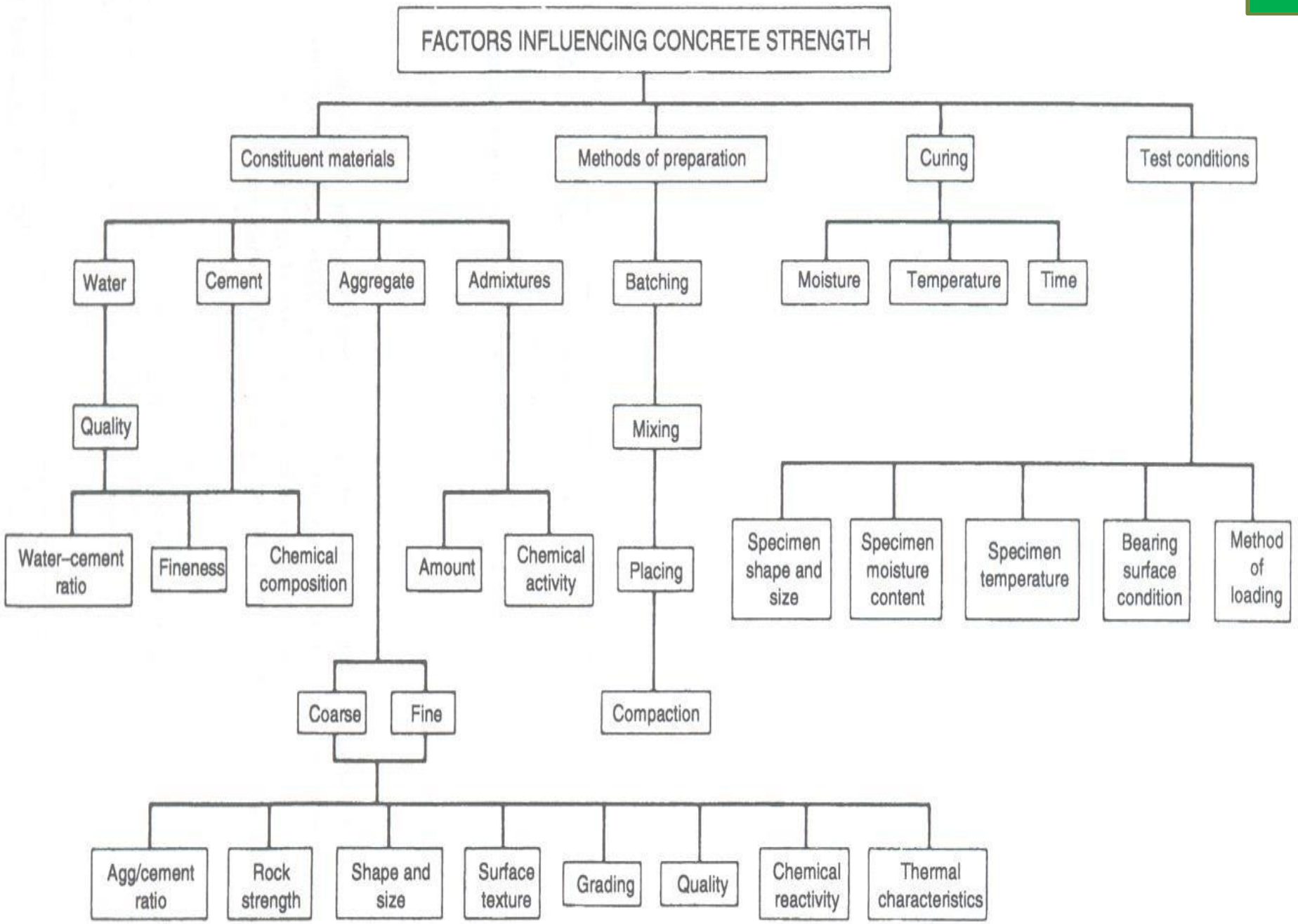




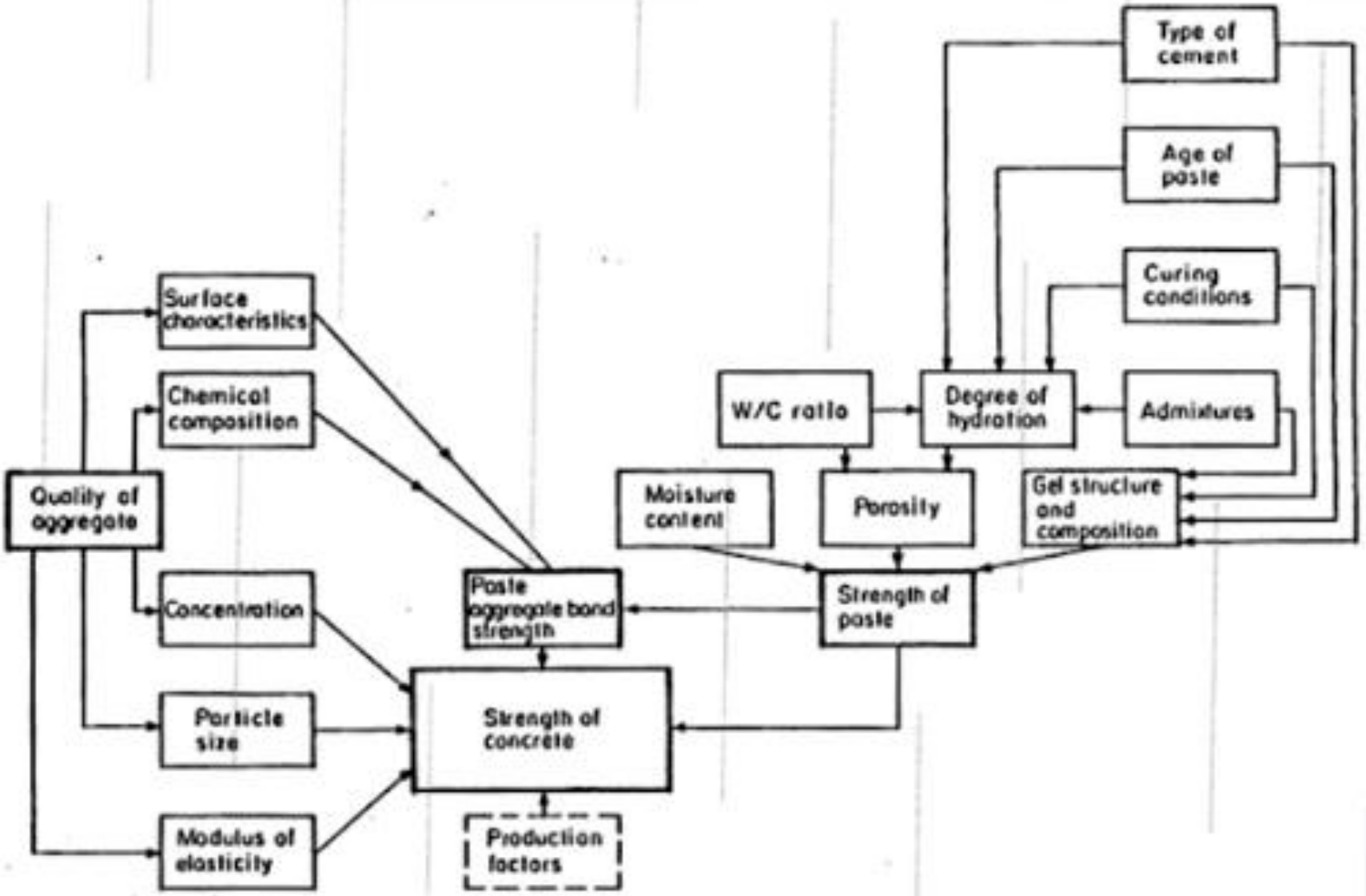
FLEXURAL TENSILE TEST



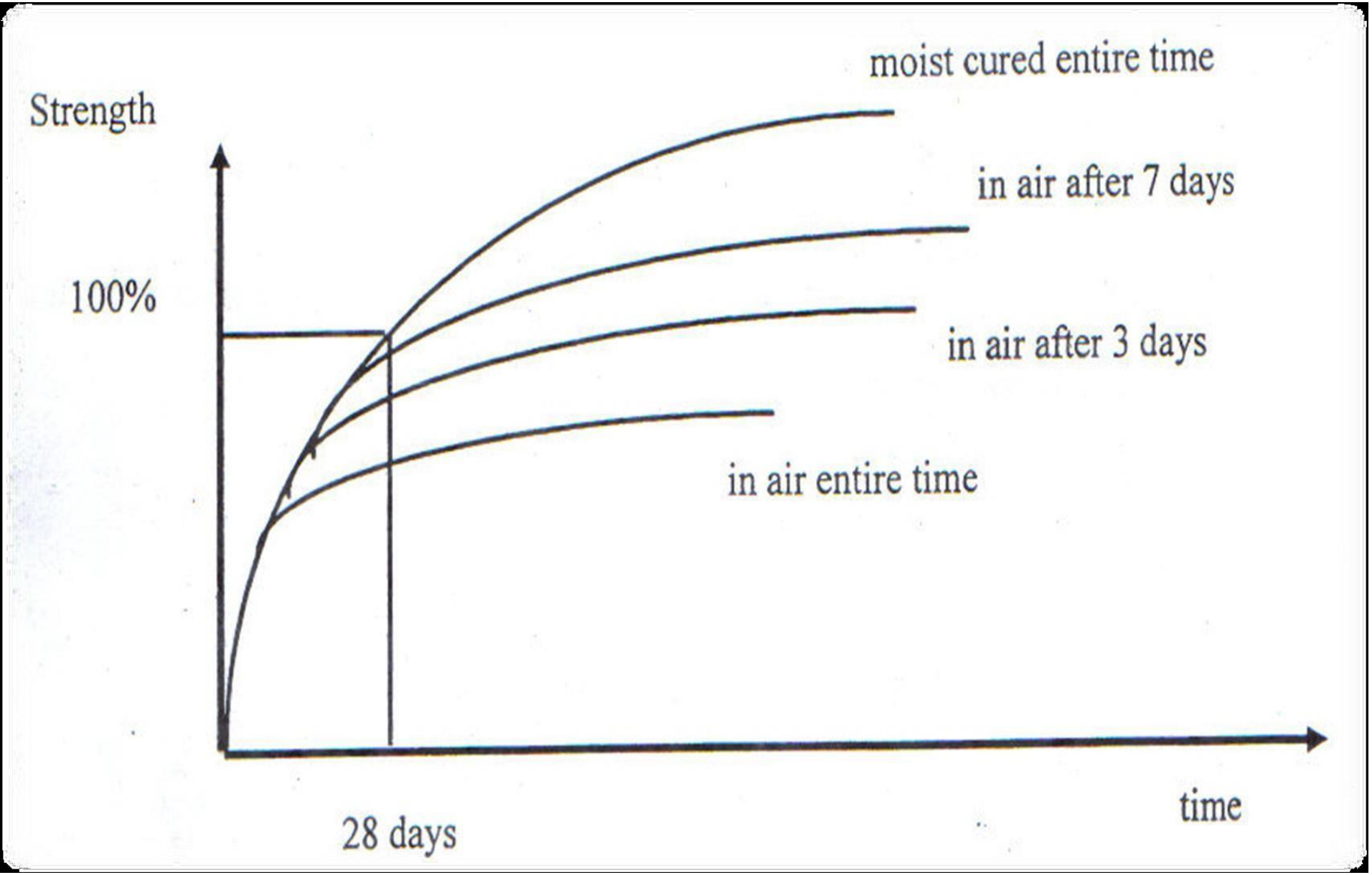



FACTORS INFLUENCING CONCRETE STRENGTH
Constituent materials
Methods of preparation
Curing
Test conditions
Water
Cement
Aggregate
Admixtures
Batching
Moisture
Temperature
Time
Quality
Mixing
Water–cement ratio
Fineness
Chemical composition
Amount
Chemical activity
Placing
Specimen shape and size
Specimen moisture content
Specimen temperature
Bearing surface condition
Method of loading
Coarse
Fine
Compaction
Agg/cement ratio
Rock strength
Shape and size
Surface texture
Grading
Quality
Chemical reactivity
Thermal characteristics

Factors Affecting Strength



Strength of concrete increase with age



Compressive Strength

Cylinder : ASTM C470

- ***Cubes : British standard 150x150x150 mm^3***

- ***Other sizes:***

  Cylinder: ***100×200 or 150×300 mm***

  Cubes: ***$100 \times 100 \times 100$ mm^3 or***

$$\sigma_c = \frac{P}{A}$$





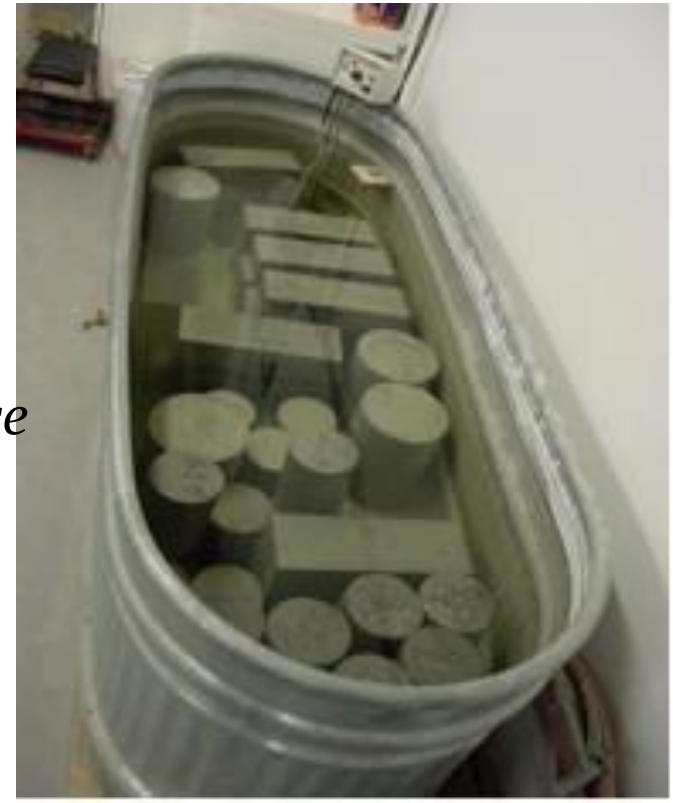


- *For 150 mm cubes fill in 3 layers compact each layer 35 times.*
- *For 100 mm cubes fill in 3 layers compact each layer 25 times.*
- *No need for capping.*









- *For 150 x 300 mm cylinder, fill in 3 layers compact each layer 25 times.*
- *Capping to obtain a plane and smooth surface (thin layer ≈ 3mm), using:*

Stiff Portland cement paste on freshly cast concrete, or mixture of Sulphur and granular material, or high-strength gypsum plaster on hardened concrete.


114
3-13-03
MD
3-13-04

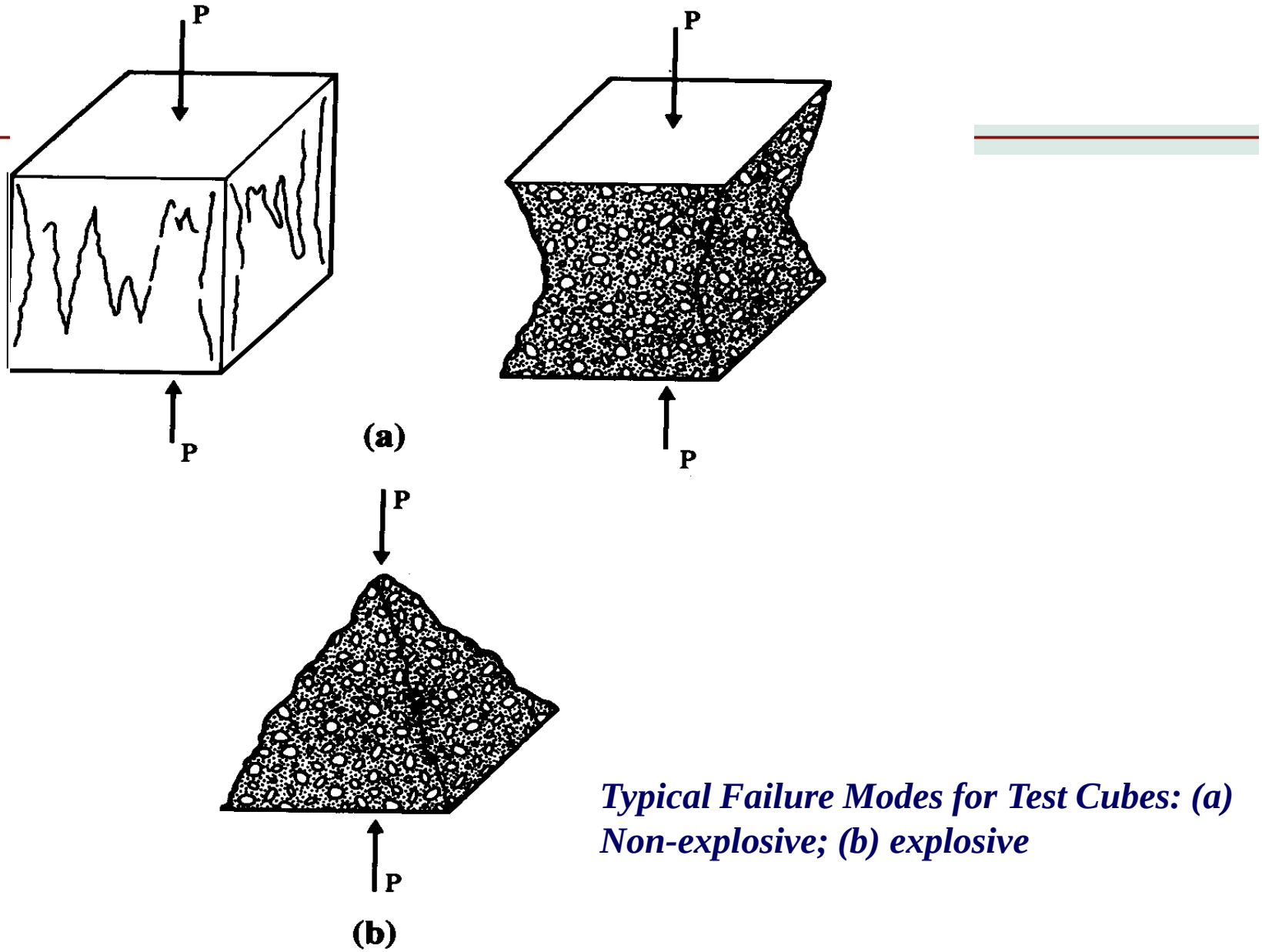





FORNEY
CAUTION
52

Factors Affecting Measured Compressive Strength

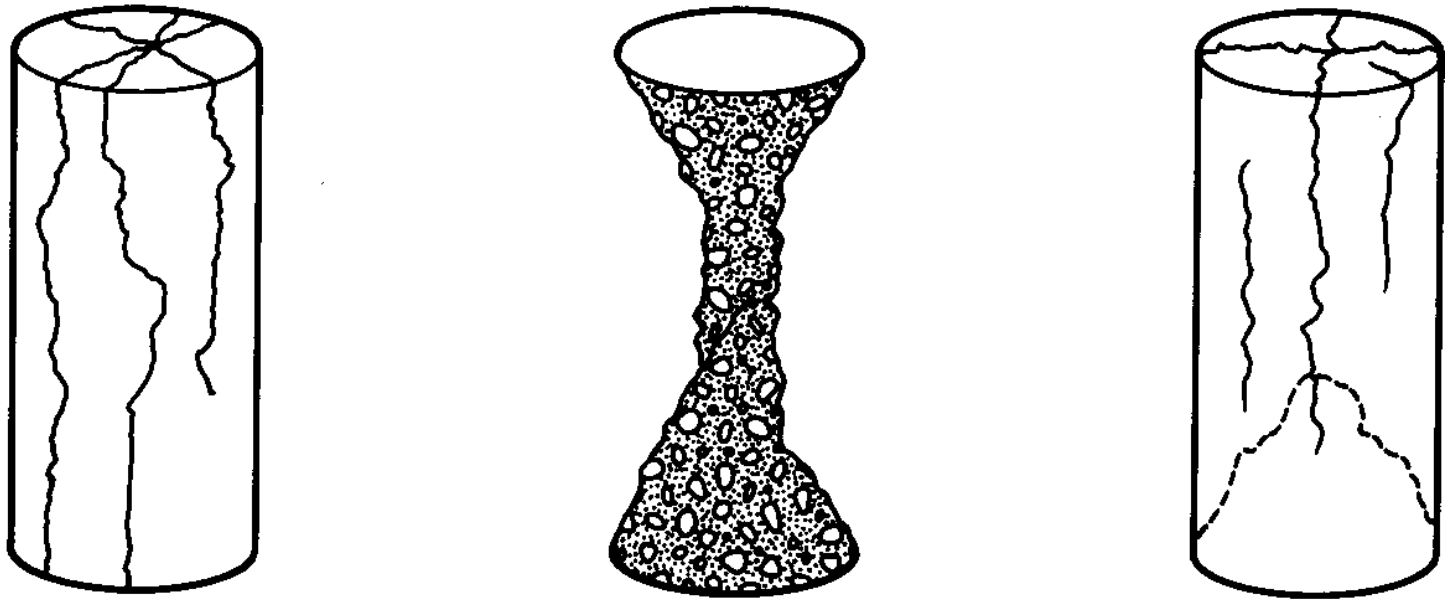
1. ***Stress Distribution in Specimens.***
2. ***Effect of L/d ratio.***
3. ***Specimen Geometry.***
4. ***Rate of Loading.***
5. ***Moisture Content.***
6. ***Temperature at Testing.***



Typical Failure Modes for Test Cubes: (a) Non-explosive; (b) explosive

Typical Failure Modes for Test Standard Cylinders:

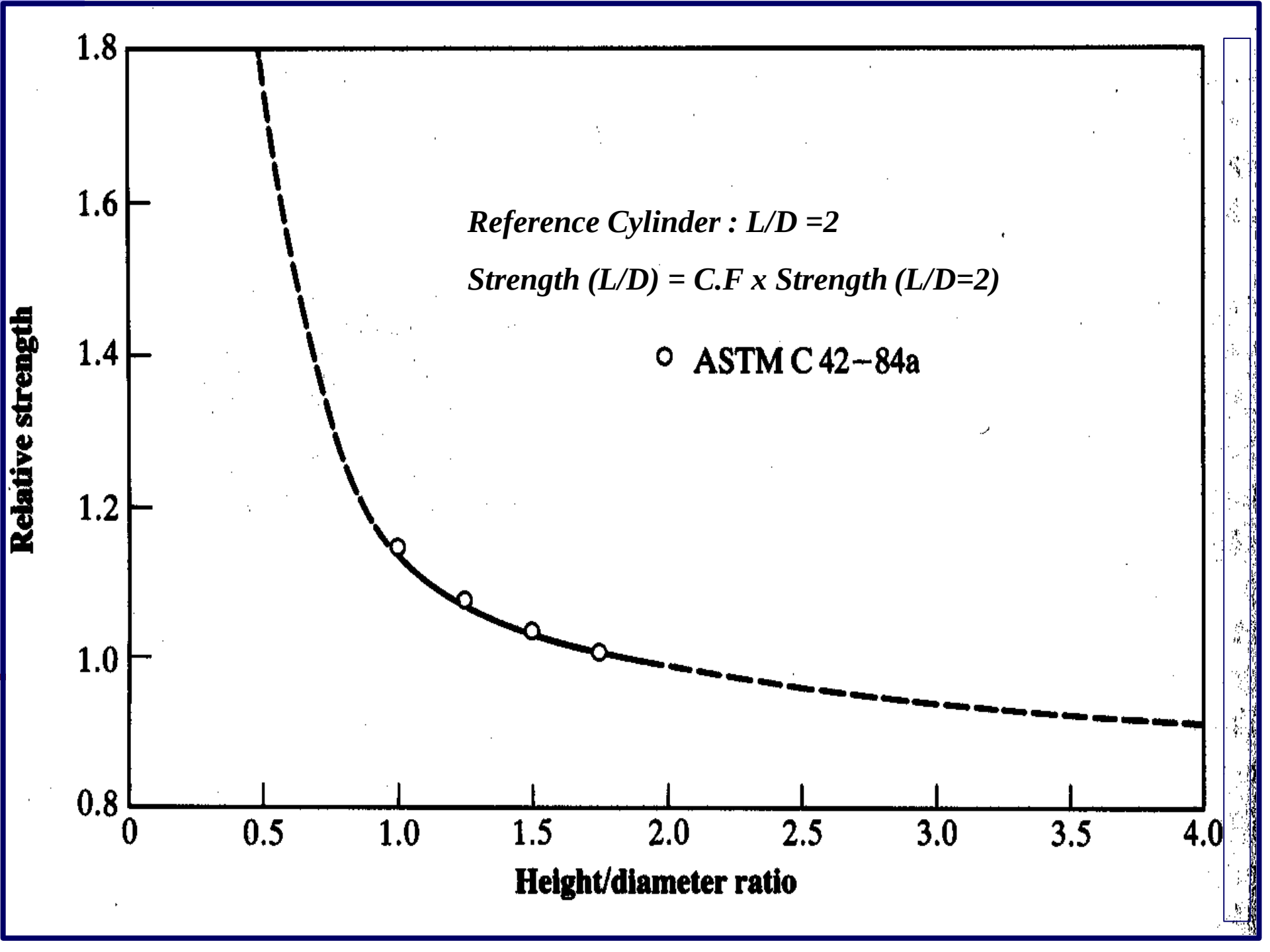
a) Splitting; *(b) Shear;* *(c) Splitting and shear (cone).*



(a) (b) (c)

2. Effect of L/d ratio

- The standard cylinder has a length to diameter ratio of 2.0
- If L/D ratio is other than 2.0 a correction factor must be applied to count for the restrainment effect of the platens


Relative strength
1.8
1.6
1.4
1.2
1.0
0.8
0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Height/diameter ratio
Reference Cylinder : L/D =2
Strength (L/D) = C.F x Strength (L/D=2)
ASTM C 42−84a

3. Specimen Geometry

- ***Different geometries for a concrete specimen can be used: Prisms, Cubes, and cylinders.***

- ***As stated before, cube are more confined by the platens thus have higher strength than cylinder made of the same concrete. It has been found that*** $\sigma_c = 1.25\ \sigma_{cyl.}$

- ***As specimen size increases, strength decreases.***

- *4. Rate of Loading*
- *Higher rate of lading → higher strength.*

5. Moisture Content

- *Standards require testing of concrete in SSD conditions (ASTM C39).*

- *6. Temperature at Testing*

Higher Temperature → lower strength

Tensile strength:

1.Direct Tensile: No standard Test

2.Indirect Tensile:

A. Splitting Tension Test.

The tensile strength of concrete is approximately equal to 10% of its compressive strength.









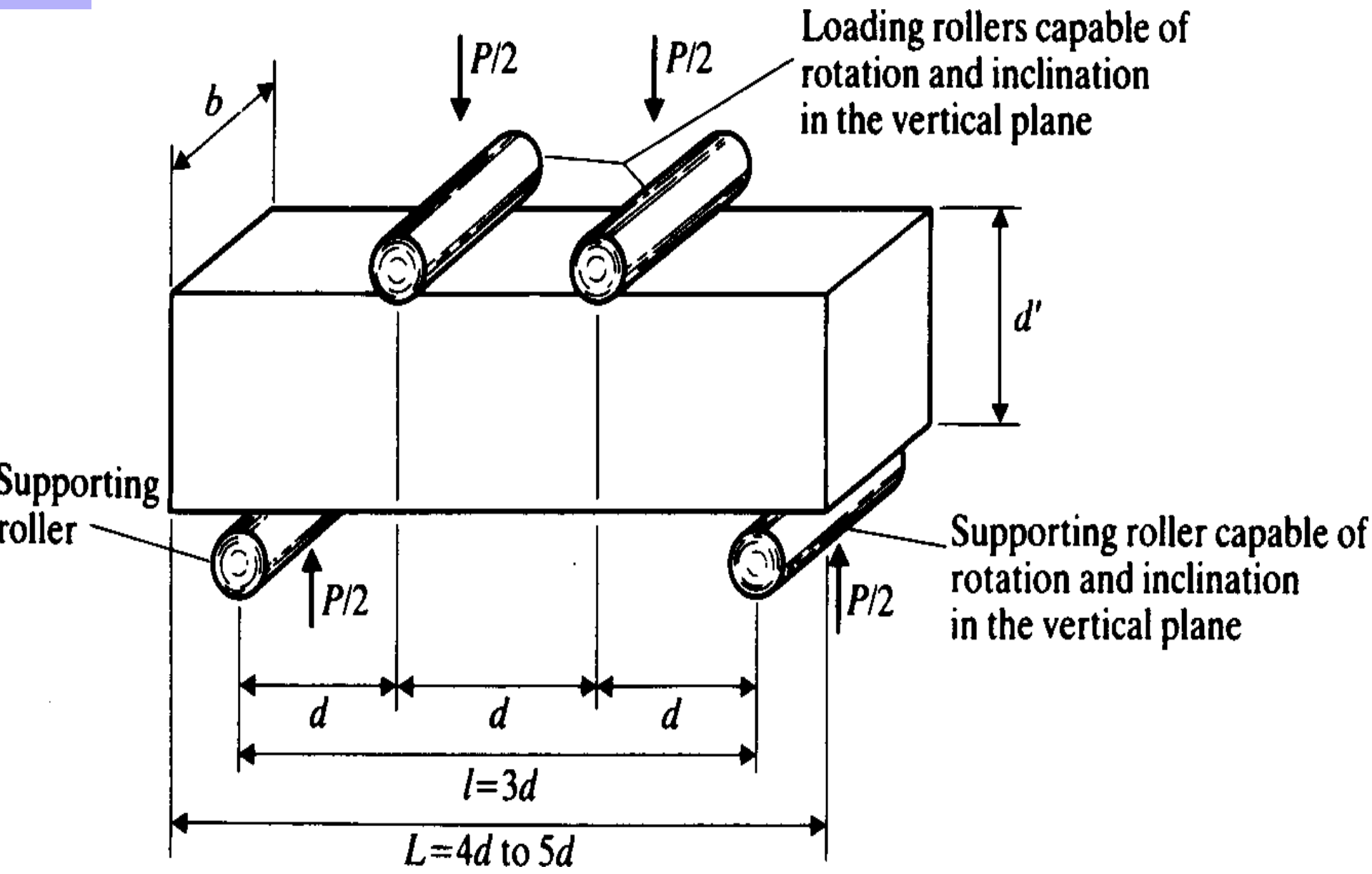
$$\sigma_{sp} = \frac{2P}{\pi LD}$$

B.Flexural strength

σ_f :

- *The test is useful since most concrete members is loaded in bending rather than in axial tension. Thus, it represents the concrete property of interest.* σ_f *is calculated as:*

$$\sigma = \frac{MC}{I}$$


Loading rollers capable of rotation and inclination in the vertical plane
P/2
P/2
b
d'
Supporting roller
Supporting roller capable of rotation and inclination in the vertical plane
P/2
P/2
d
d
d
l=3d
L=4d to 5d




ELE



- This test is mostly used for quality control of highways and airport runways. It gives more useful information than do compression tests.

Flexural strength:

Affected by:

- Specimen Size ↑ → strength ↓
- Temperature: Same as in compression.

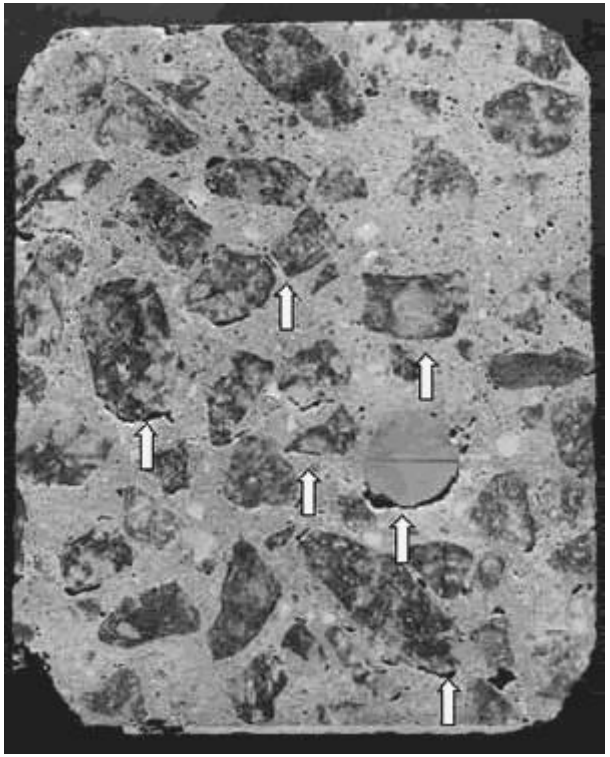
The tensile strength of concrete is approximately equal to 10% of its compressive strength.

Strength of concrete

Strength = ability to resist stress without failure.

Concrete strength made of:

1. *Strength of paste or mortar.*
2. *Strength of CA-paste (mortar) interface.*
3. *Strength of CA.*



Cracks at the bond between the aggregate, rebar, and paste (see arrows).

Factors Affecting Strength of Concrete

1.Water/Cement Ratio

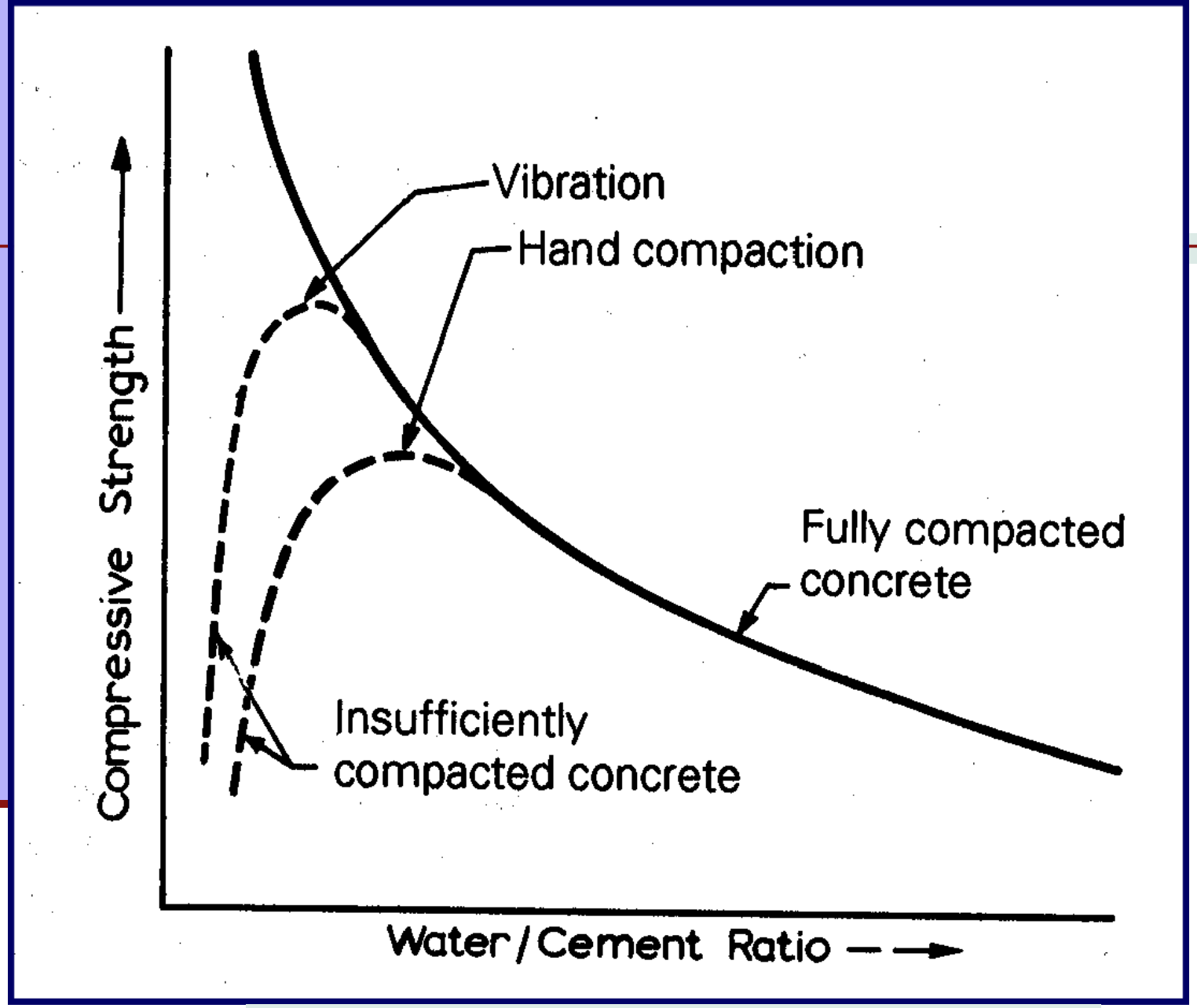
Since the W/C ratio controls the porosity of concrete, it controls the strength as well.

W/C ↑ → *strength* ↓

2.Degree of

Compaction Strength =

f (full compaction)



Relation between strength and W/C ratio

3. Curing Time:

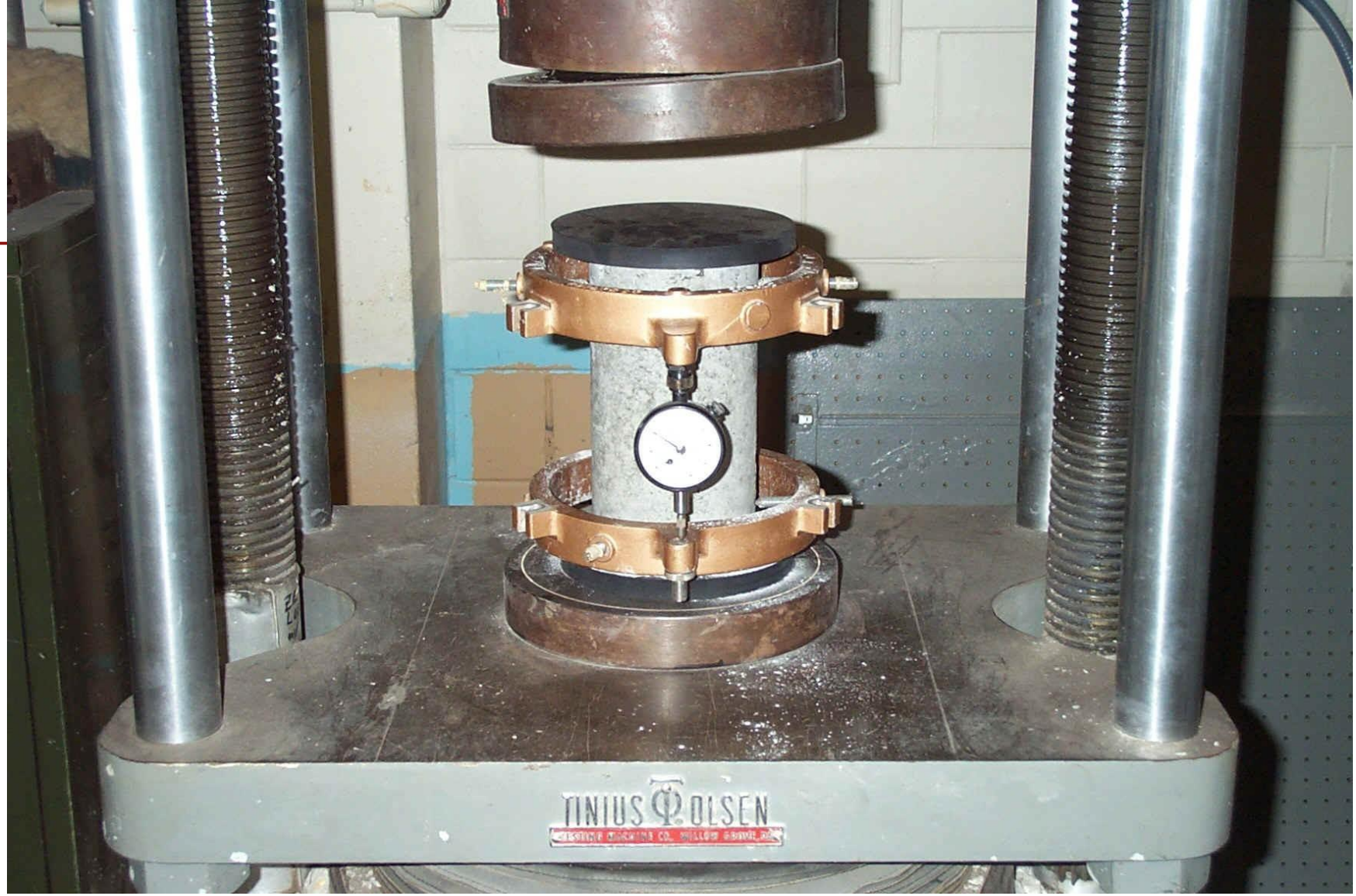
- In practice, it is common to obtain 7-day as well as 28-day compressive strength.

4. Cement:

- The effect of Portland cement on concrete strength depends on the chemical composition and fineness of the cement.

5. Aggregates: aggregates Shape and Texture

Texture depends on whether aggregate is natural (gravel)or crushed.



- Stress-Strain Diagram of Concrete:

FACTOR INFLUENCING THE STRENGTH OF CONCRETE

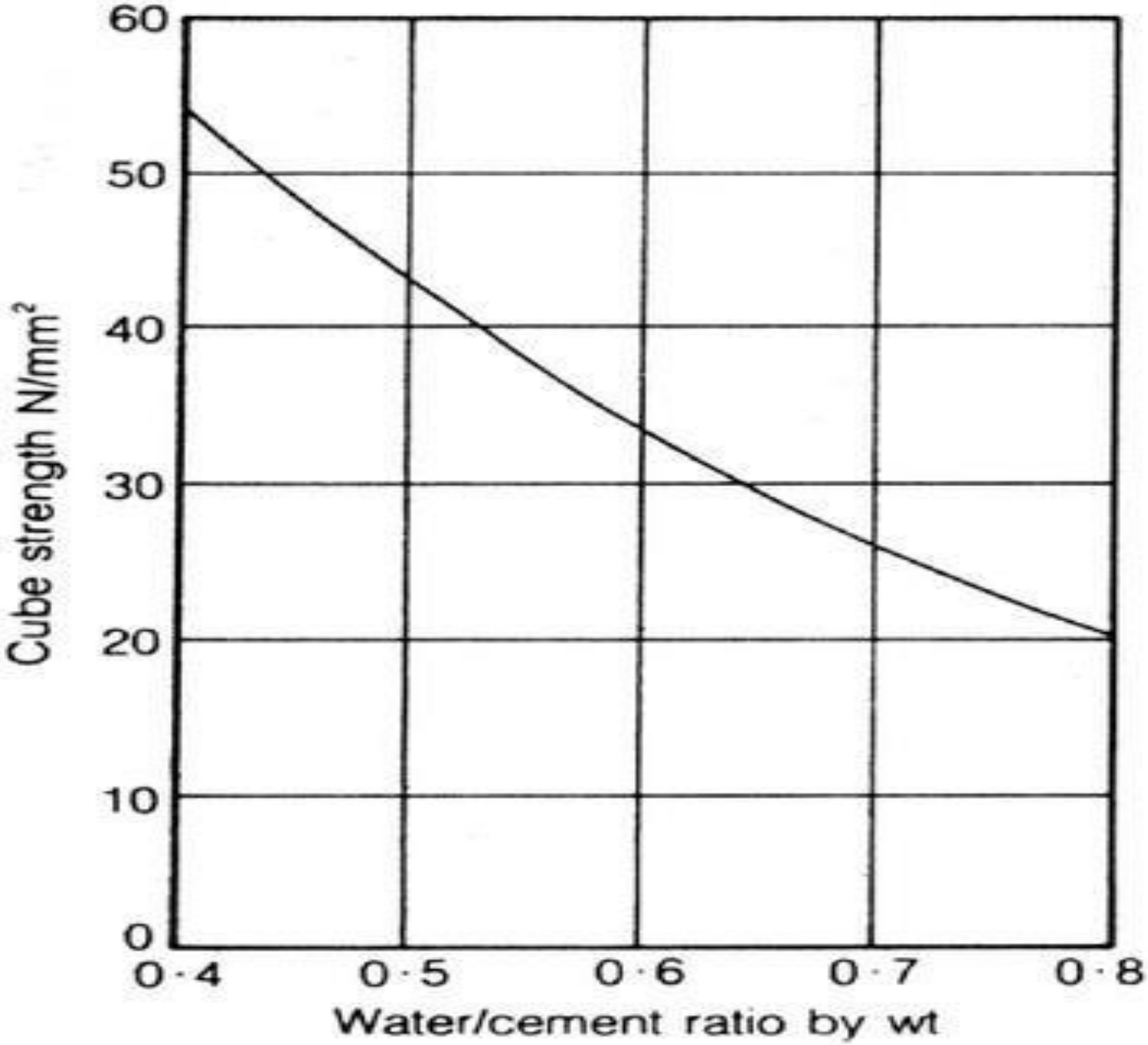
1. Influence of the constituent materials.

a) Cement

- Fineness of cement increase strength.
- Chemical composition.
- Type of cement (eg. RHPC, SRPC, LHPC).

b) Water

- Water cement ratio (w/c) required for hydration process.
- The lower w/c, the greater is the compressive strength and vice versa.


60
50
40
30
20
10
0
Cube strength N/mm²
0·4
0·5
0·6
0·7
0·8
Water/cement ratio by wt

c) Aggregate

- Bond strength between aggregate influence by surface texture, shape and cleanliness.
- The compressive strength of concrete tends to increase with the decrease in the size of coarse aggregate. This due to the facts that smaller size aggregates provide larger surface area for bonding with the mortar matrix.
- Aggregate surface roughness has a considerable affect on bond strength (greater the roughness, higher is the bond strength) due to improvement in mechanical interlocking.

d) Admixture

- Effect of particular admixture (eg. accelerator, retarder, plasticizer etc.) depend on the precise nature of the admixtures themselves.

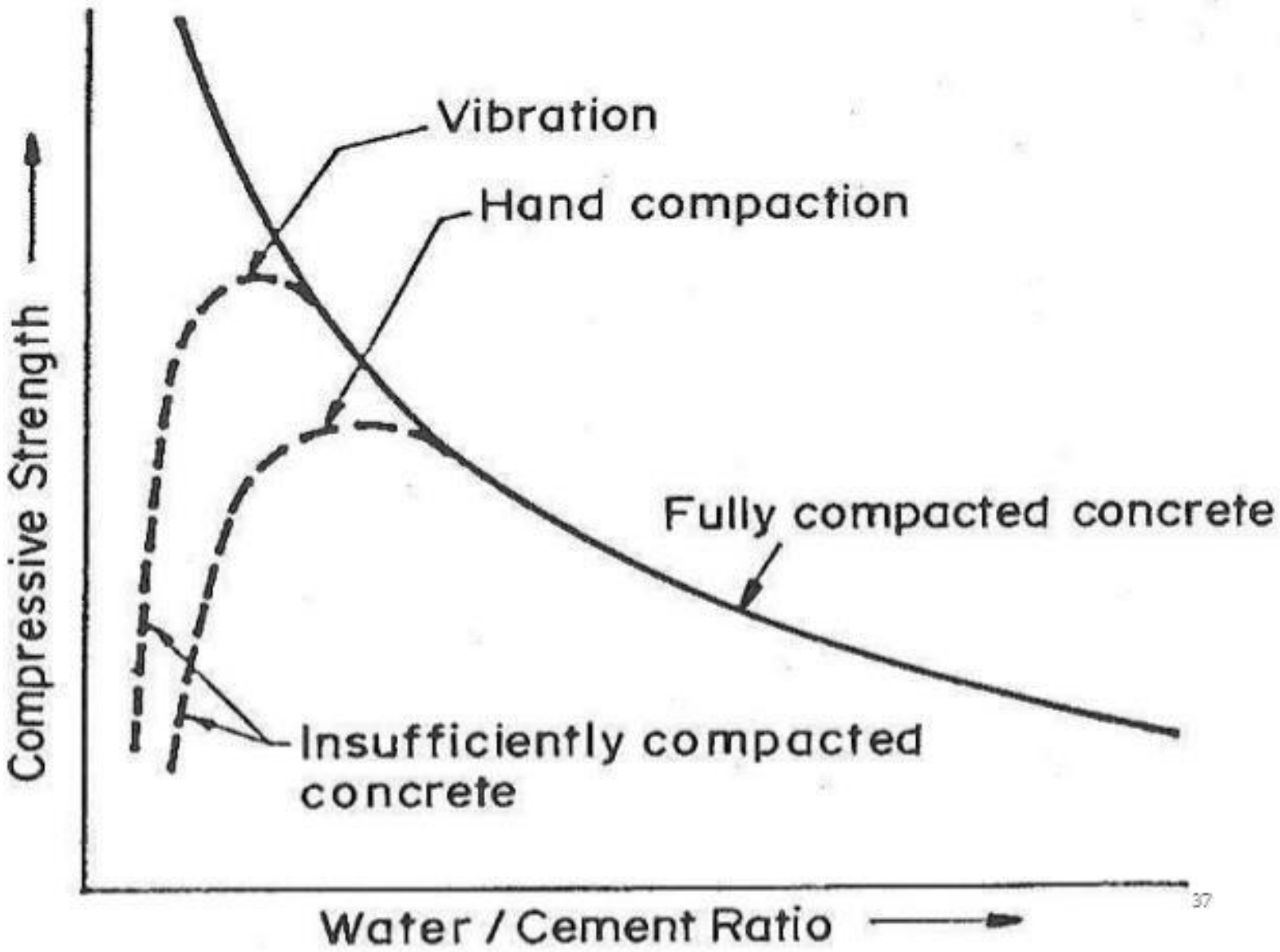


2. Degree of Compaction

- Proper compaction increase strength.

- When the concrete compacts, it has a very low porosity, thus result in a very high strength.

- The increase in the strength of concrete is probably influenced by the volume of voids in concrete i.e. entrapped air, capillary pores, gel pores or entrained air.


Vibration
Hand compaction
Fully compacted concrete
Insufficiently compacted concrete
Compressive Strength
Water / Cement Ratio

3. Influence of curing.

- Curing is used for promoting the hydration of cement and consists of a control of temperature and moisture movement from and into the concrete.
- The longer period of curing, the greater its final strength.

- Early-age: ages less than 7days
- Later-age: ages exceeding 28days.



4. Influence of test conditions

- Specimen shape and size – cube, cylinder and square prism.
- Specimen moisture content and temperature.
- Method of loading.

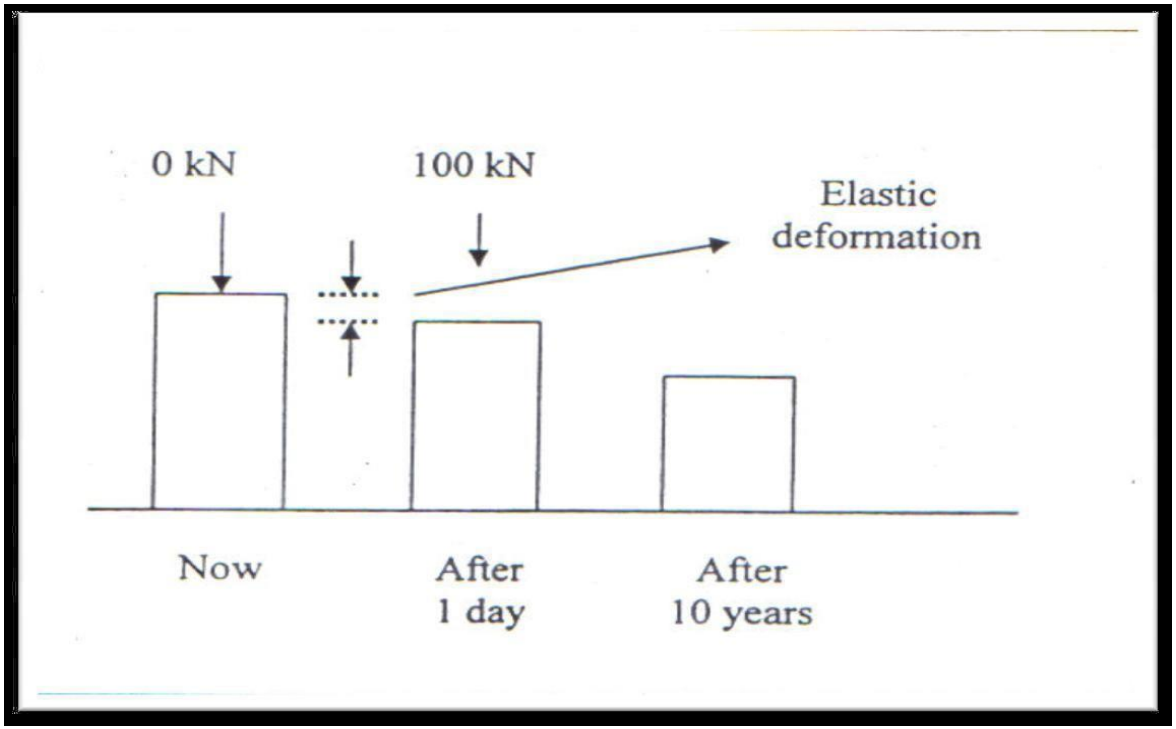


2. DEFORMATION UNDER LOAD

- It is a stress strain relationship under normal loading and under sustained loading.
- Under normal loading: the first effect of applying a load to concrete is to produce an elastic deformation i.e. as the load increases deformation increases.
- Under sustained loading: the continue application of stress causes a slow deformation known as creep.
- The increase of deformation is not proportional , as the time passes the deformation is lesser.

Elastic deformation:

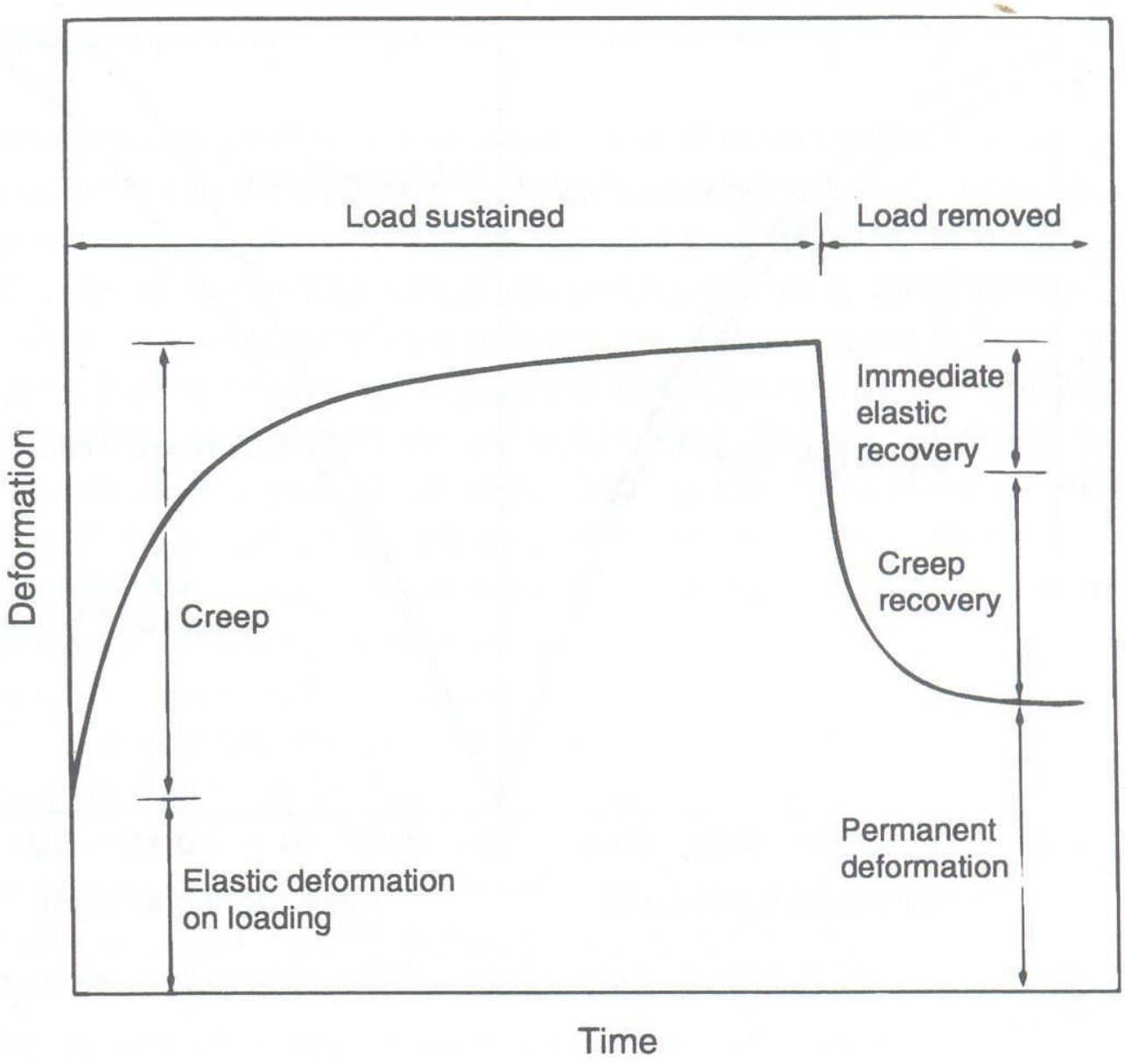
- When the applied load is released, the concrete does not fully recover its original shape.
- Under repeated loading and unloading, the deformation at a given load level increases.



Modulus of elasticity:

- Defined as the ratio of load per unit area (stress) to the elastic deformation per unit length (strain).
- The modulus of elasticity for most concretes at 28 days, ranges from 15 – 40 kN/mm^2.

$$E = \frac{stress}{strain} = \frac{\sigma}{\varepsilon}$$



A typical illustration of deformation of concrete subjected to constant load.

3. DURABILITY

- Defined as its resistance to deterioration processes that may occur as a result of interaction with its environment (external) or between the constituent materials or their reaction with contaminants present (internal).

- Ability to with stand the damaging effects of the environment over a long period of time.

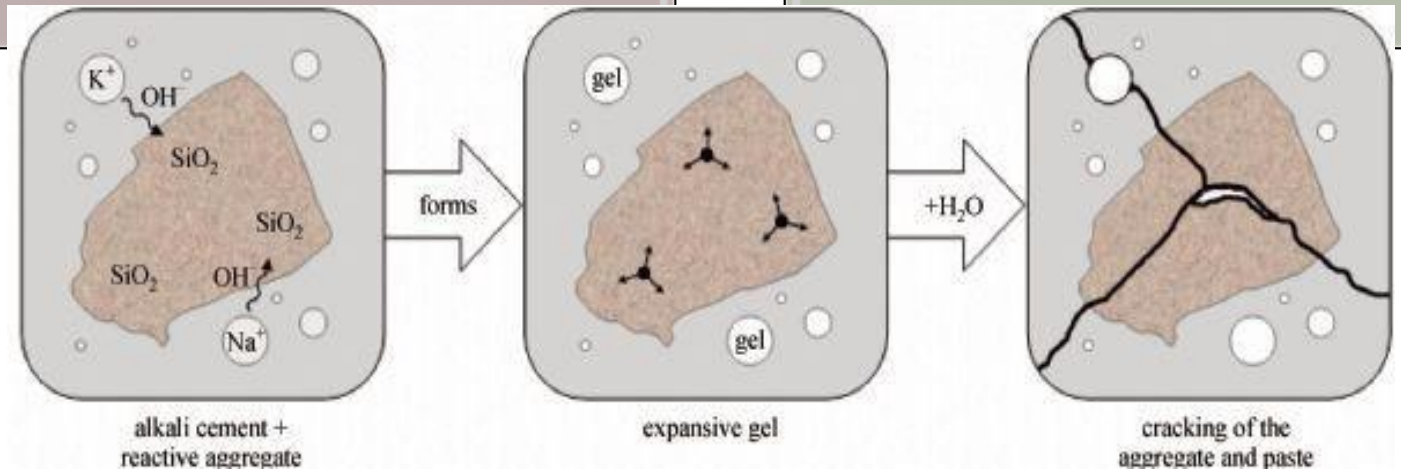
- The absence of durability maybe caused either by the environment to which the concrete is exposed i.e. external or internal causes.

INTERNAL CAUSES

- The alkali-aggregate reaction, volume changes due to the differences in thermal properties of aggregate and cement paste and the permeability of the concrete.

EXTERNAL CAUSES

- physical, chemical and mechanical
- weathering, occurrence of extreme temperature, abrasion, electrolytic action.
- The common forms of chemical attack : leaching out of cement and action of sulphates



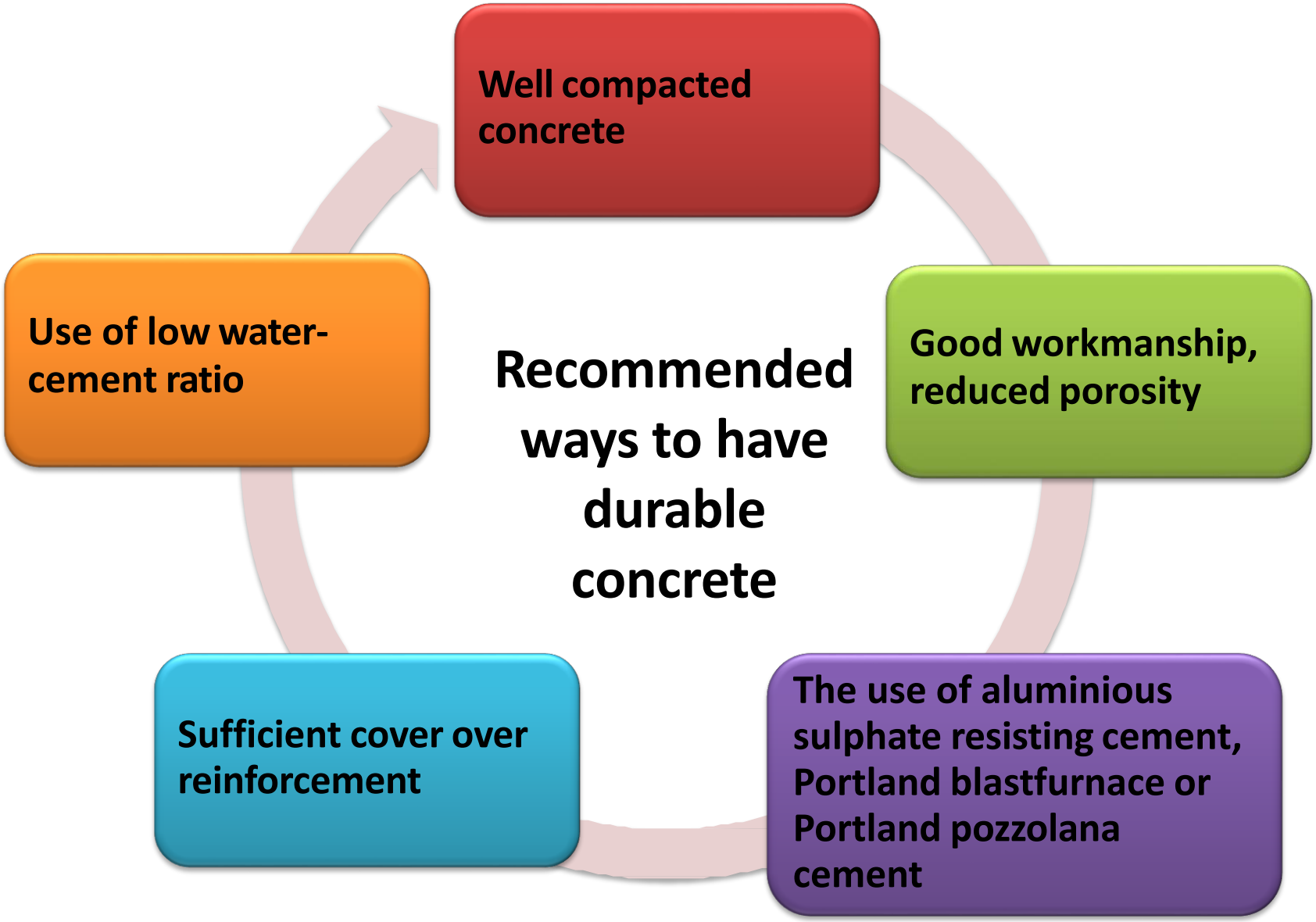
Leaching is the dissolution of some concrete components in a liquid. For example, lime is dissolved according to this reaction : Ca(OH)2 > Ca2+ + 2 OH- After leaching, concrete porosity is increased and its deterioration is enhanced

LEACHING IN CONCRETE







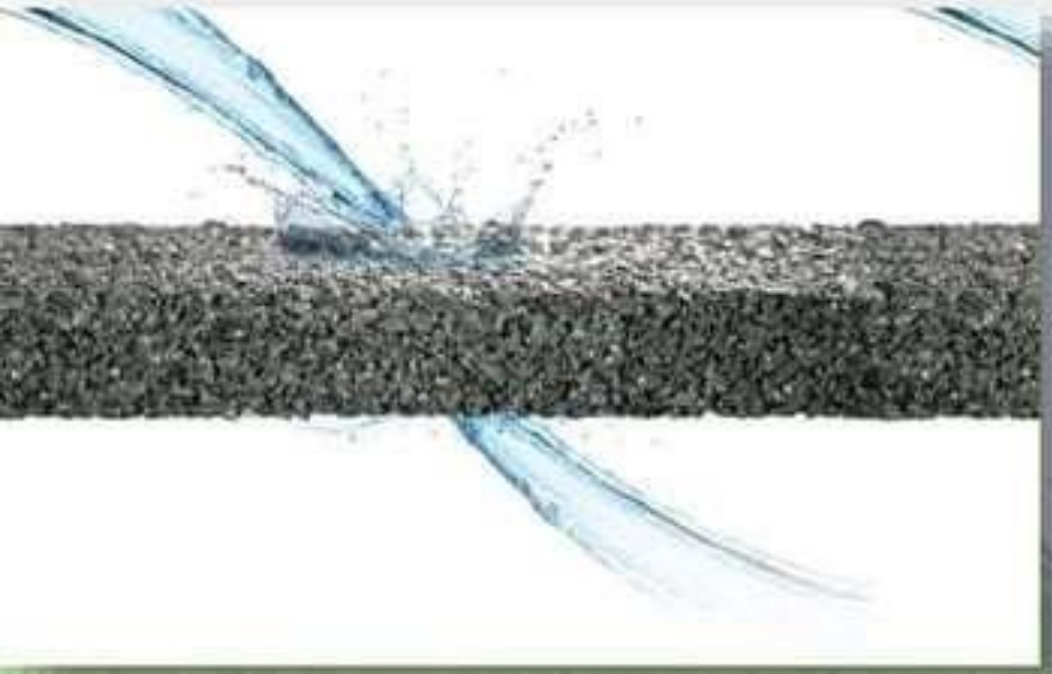

Recommended ways to have durable concrete
Well compacted concrete
Good workmanship, reduced porosity
The use of aluminious sulphate resisting cement, Portland blastfurnace or Portland pozzolana cement
Sufficient cover over reinforcement
Use of low water-cement ratio

4. PERMEABILITY

- **Concrete has a tendency to be porous due to the presence of voids formed during or after placing.**
- Penetration by substance may adversely affect durability e.g. $Ca(OH)_2$ leaches out.
- Ingress of air and moisture resulting in corrosion.
- Important with regards to water tightness of liquid retaining structure.

What is permeability?

Permeability is **defined** as the property that governs the rate of flow of a fluid into a porous solid.





PERMEABILITY OF CONCRETE

- For completing hydration of cement about 38 % of water by weight of cement is required to fill the gel pores.
- If more than 38 % of water is used, than excess water will cause undesirable capillary cavities and the concrete becomes porous. Porous concrete has a higher permeability





Pores in cement paste

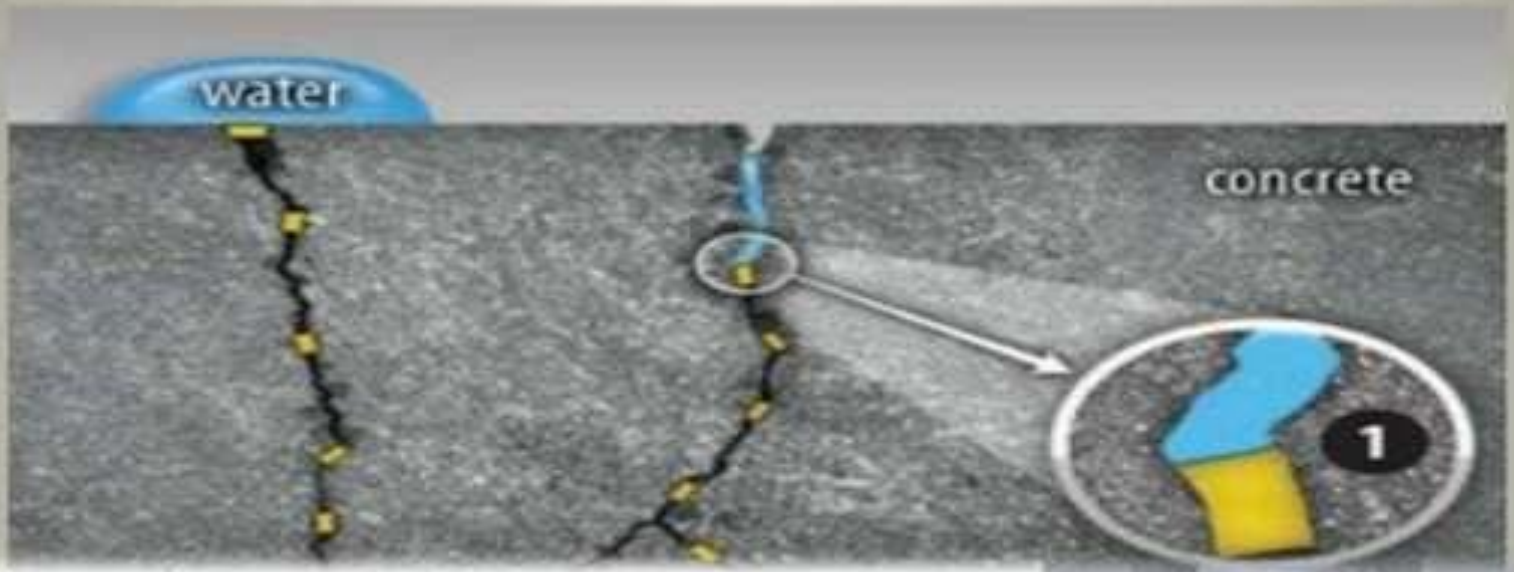
A. Gel pores

B. capillary pores

- **Gel pores :** it is so small that hardly any water can pass through under normal condition.
- Its permeability is 7×10^{-16} m/s which is about 1/100 of that paste hence it do not contribute to permeability
- **Capillary pores :** it consist about 0 to 40 % of the paste volume.
- They are much larger than gel porous



Capillary Pores in Concrete



C² Lithium Technology

Concrete Porosity

- Capillary Pores 0.1µ
- Gel Pores <0.01µ



University of Aberdeen

Kings College, Aberdeen, UK

Importance of Permeability

- In reinforced concrete, ingress of water and air will result in corrosion of steel leading to expansion, cracking, and disruption of concrete.



- The penetration of deleterious material in solution may adversely affect the durability of concrete. $\underline{Ca(OH)_2}$ leaches out and aggressive liquids attack the concrete.

CALCIUM COMPOUNDS

Slaked lime (Hydrated lime)

- **Calcium hydroxide**, $Ca(OH)_2$
- It is a colorless crystal or white powder and is obtained when calcium oxide (quicklime) is mixed, or slaked with water.
- Used in many applications, including food preparation.







- If concrete becomes saturated with water due to permeability, it is more vulnerable to <u>frost action</u>.



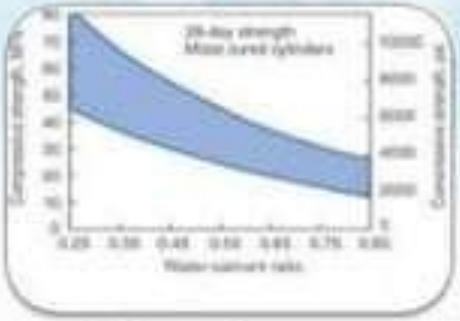
- ✓ The permeability is very important in case of liquid retaining structures like water tanks and dams where water-tightness is necessary.





FACTORS AFFECTING PERMEABILITY

The main factors affecting permeability are:



Water/ cement ratio



Properties of cement



Aggregate



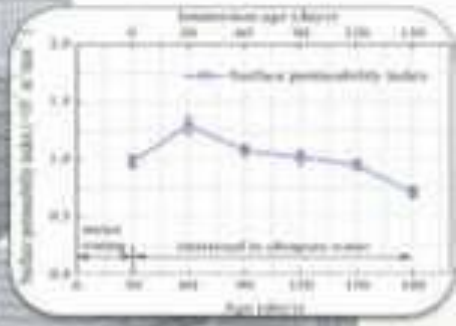
Absorption and homogeneity of concrete



Curing



Use of admixtures



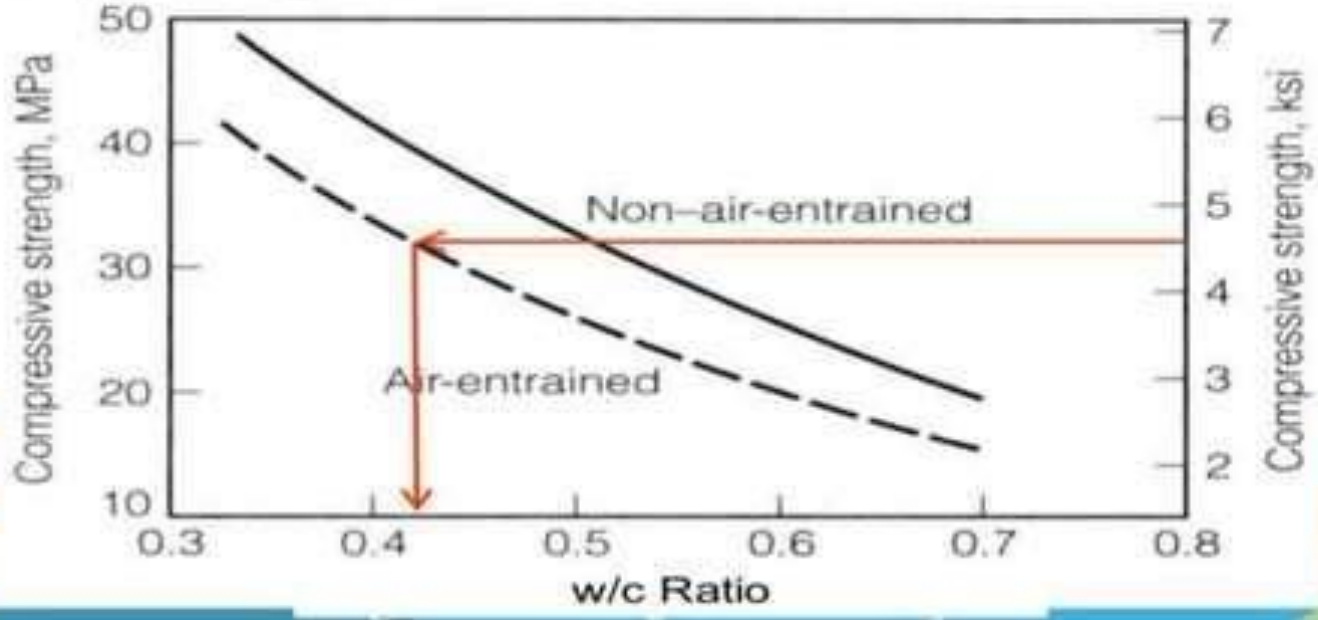
Age of concrete

WATER/ CEMENT RATIO:

- The permeability is lower with lower water/ cement ratio & with higher cement content permeability becomes less.

WATER-CEMENT RATIO

historical records of strength are used to plot f'_c vs. w/c



Properties of Cement:

- For the same water/ cement ratio, coarse cement tends to produce a paste with higher porosity of cement than a finer cement. In general, higher the strength of cement paste, the lower will be the permeability

coarse cement



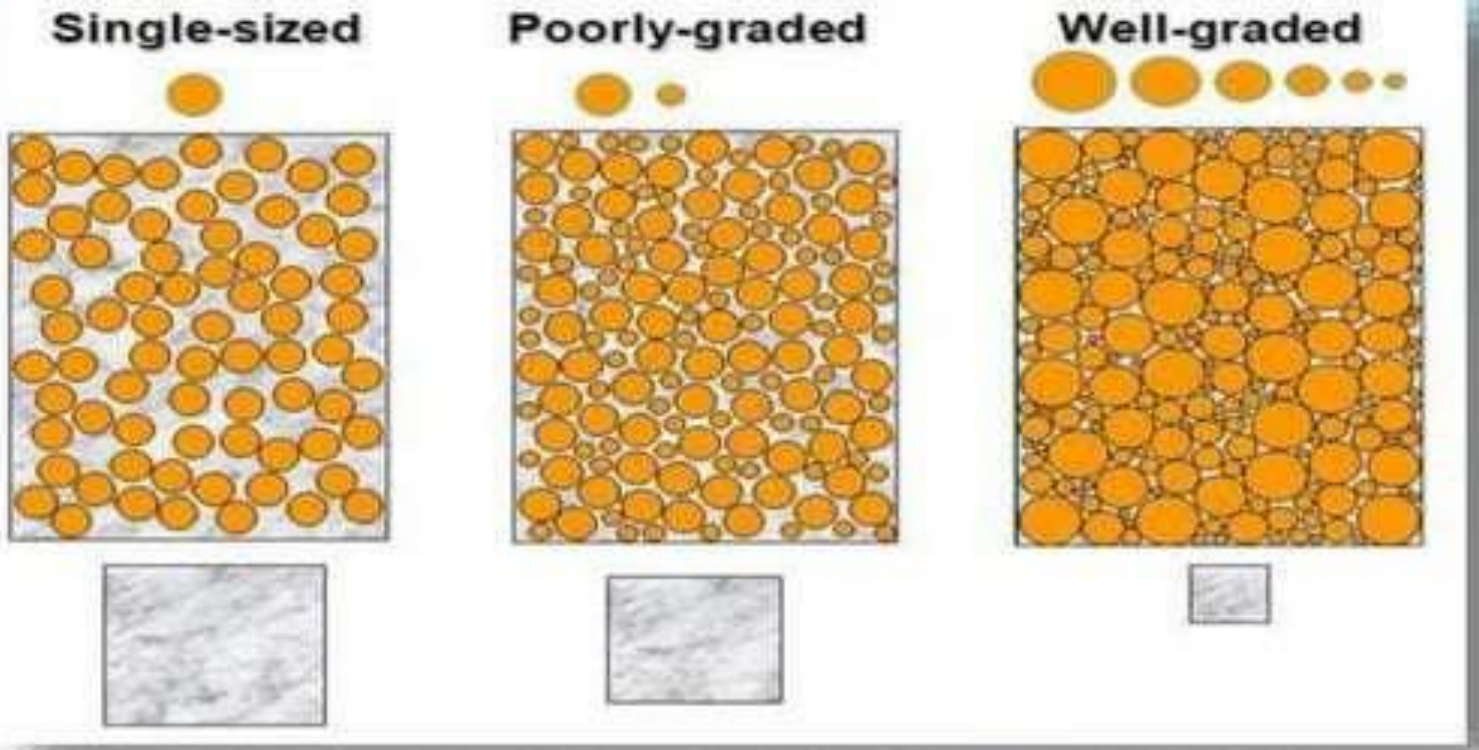
finer cement



Aggregate:

- For a given water/ cement ratio, greater the maximum size of aggregate greater is the permeability. This is because of the relatively larger voids. Well graded aggregate reduces the permeability




Single-sized
Poorly-graded
Well-graded



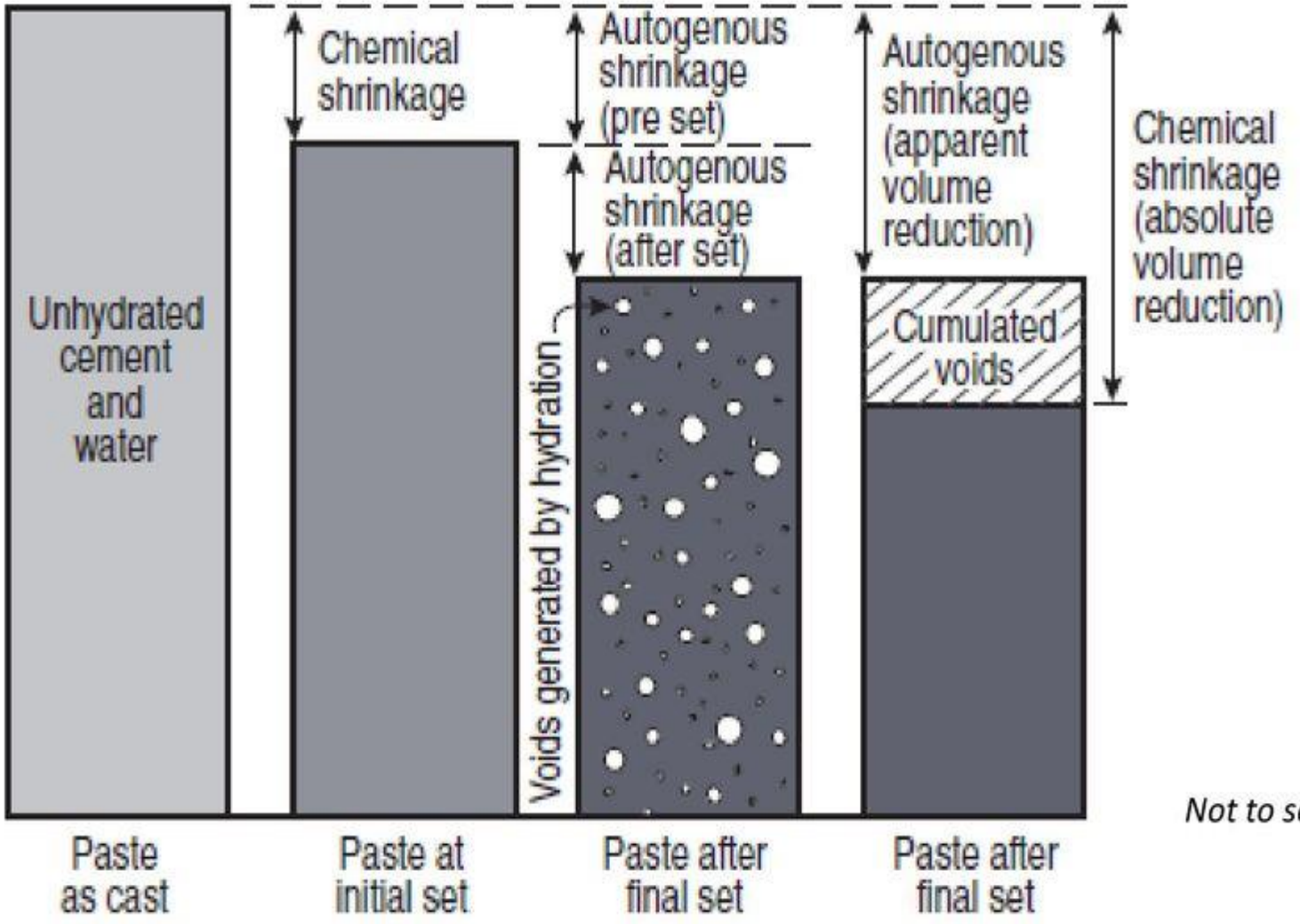
Factors influencing permeability are

Factor	Description
Water-Cement Ratio (W/C)	A higher water-cement ratio increases the porosity of concrete, leading to higher permeability.
Curing Conditions	Proper curing improves the hydration process, reducing voids and permeability. Poor curing increases permeability.
Aggregate Size and Quality	Poorly graded or weak aggregates can lead to voids in the concrete mix, increasing permeability.
Cement Content	Low cement content can lead to higher porosity and permeability. Higher cement content can reduce the amount of voids.
Compaction	Insufficient compaction can result in voids or air pockets, leading to increased permeability.
Admixtures	Certain admixtures, like water reducers or pozzolans (e.g., fly ash, silica fume), can reduce permeability by refining pores.
Concrete Age	Older concrete tends to be less permeable due to ongoing hydration and reduction of capillary pores over time.
Cracks and Micro-cracking	Cracks caused by load, shrinkage, or temperature can drastically increase the permeability of concrete.
Type of Cement	Different types of cement (e.g., Portland, sulfate-resistant) can have varying effects on the permeability of concrete.
Temperature	High temperatures during mixing and curing can lead to rapid evaporation of water, increasing permeability.

5. SHRINKAGE

- Caused by the settlement of solids and the loss of free water from the plastic concrete (plastic shrinkage), by the chemical combination of cement with water and by the drying of concrete (drying shrinkage).
- The shrinkage is dependent on the amount of drying that can take place.
- Influenced by the humidity and temperature of the surrounding air, the rate of air flow over the surface and the proportion of the surface area to volume of concrete

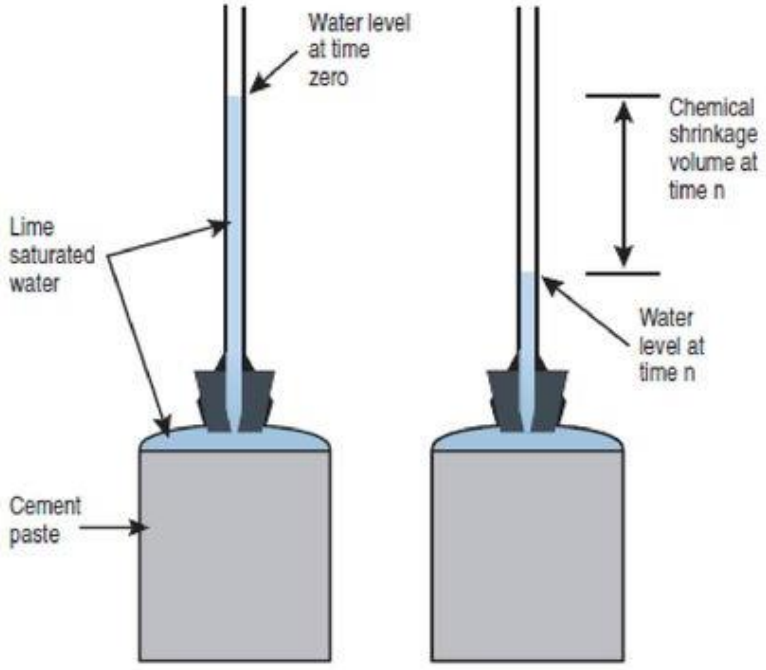
Early Age Volume Changes

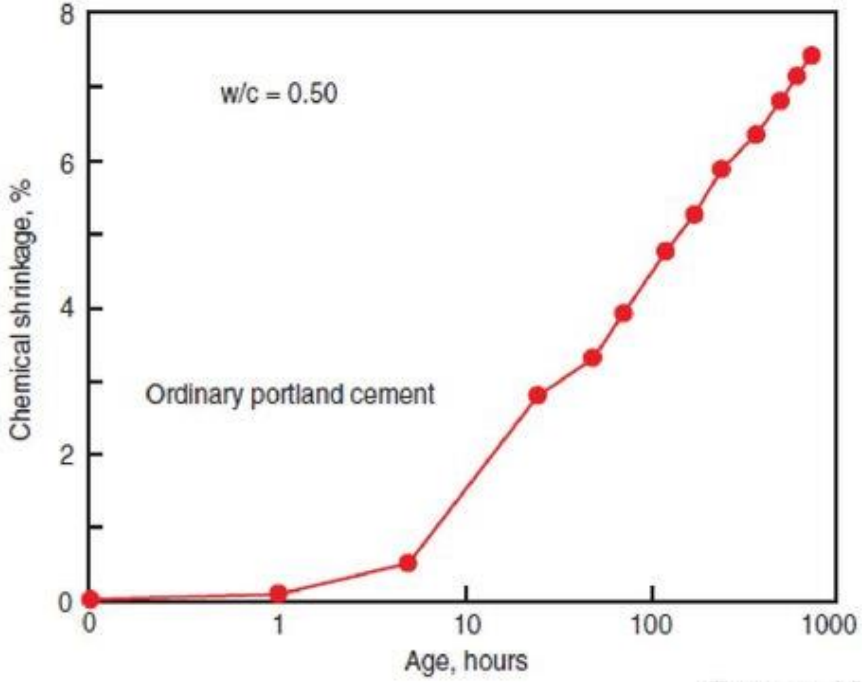


Not to scale.

Chemical Shrinkage

ASTM C1608- *Standard Test Method for Chemical Shrinkage of Hydraulic Cement Paste*



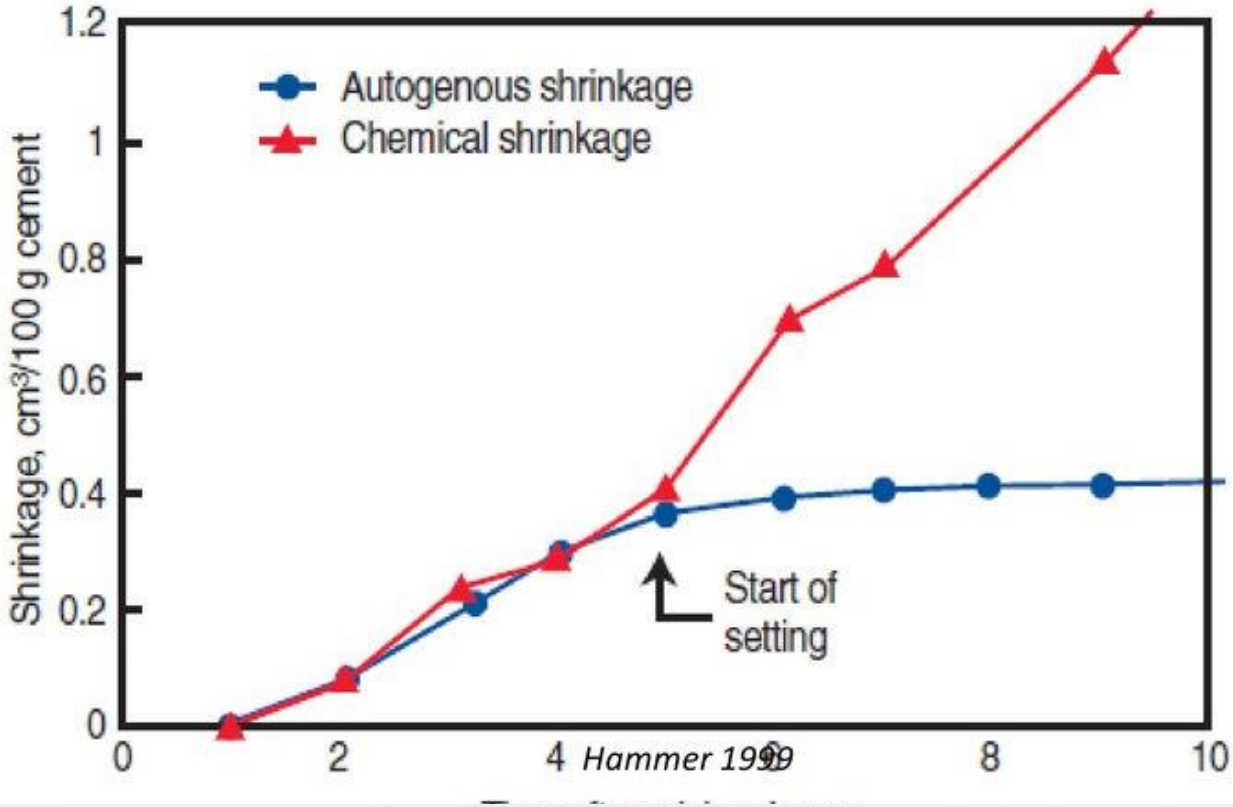


Tazawa 1999

- volume of the hydration product is smaller than volume of un-hydrated cement and water (0.06Ch) & is chemical shrinkage

Autogenous Shrinkage

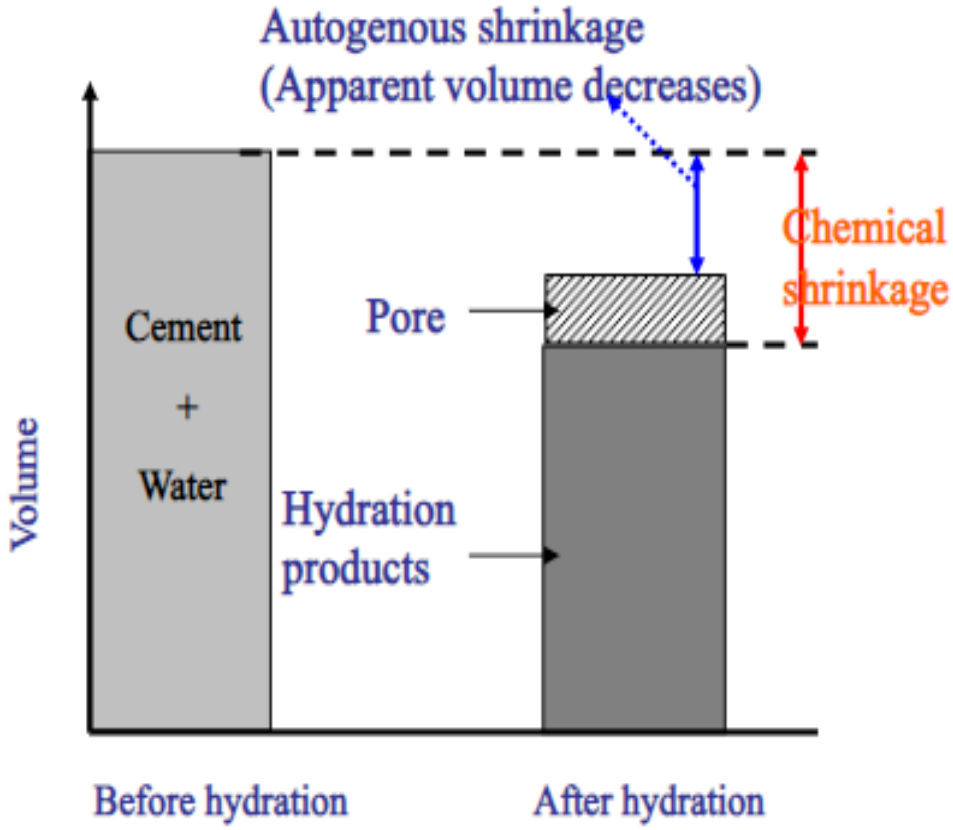
ASTM C1698- *Standard Test Method for Autogenous Strain of Cement Paste and Mortar*



Autogenous shrinkage is a macroscopic reduction in length under constant temperature and without any moisture migration to or from the concrete.

Autogenous shrinkage -mechanics

Root reason: the total volume decreases during cement hydration process (chemical shrinkage) .



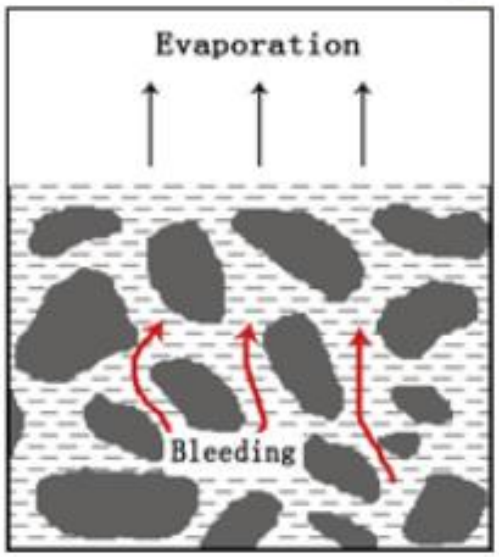
- Chemical shrinkage is about: 6.4×10^{-2} mL/g ;
- Autogenous shrinkage is one of the manifestations of chemical shrinkage, chemical shrinkage equals to the sum of autogenous shrinkage and pore volume formed

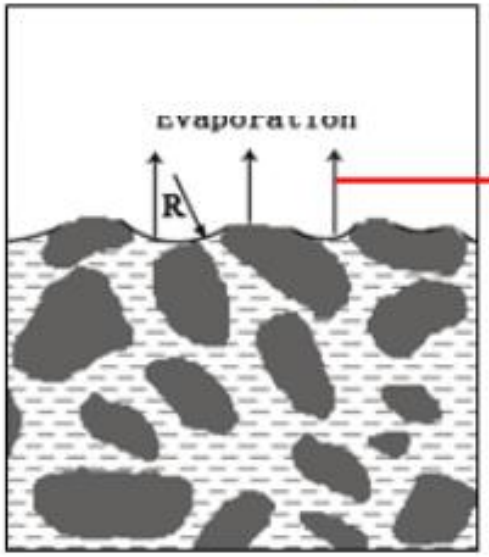
types of shrinkage:

a) Plastic Shrinkage

- Shrinkage which takes place before concrete has set.
- Occurs during the first few hours after fresh concrete is placed.
- During this period, moisture may evaporate faster from the concrete surface than it is replaced by bleed water from lower layers of the concrete mass.

Mechanics of plastic shrinkage and cracking



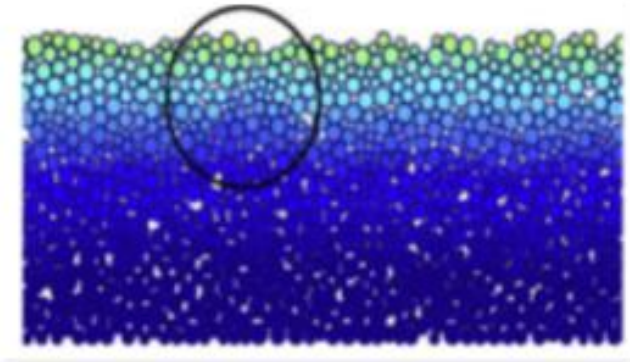


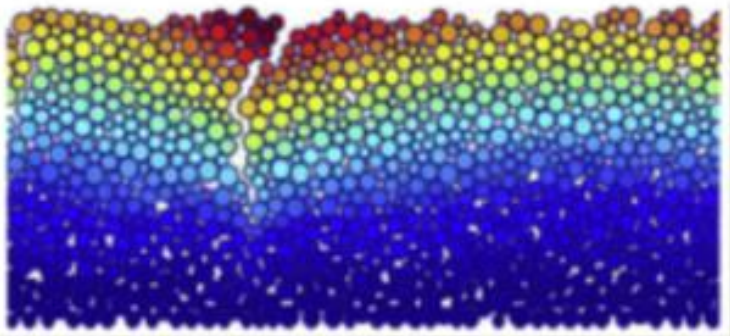
$$\Delta P = \frac{2\gamma}{r}\cos\theta$$

The shrinkage driving force is greater than the tensile strength between particles

Bleeding rate ≧ Evaporation rate

Bleeding rate < Evaporation rate





b) Drying Shrinkage

- When a hardened concrete, cured in water, is allowed to dry it first loses water is drawn out of its cement gel.

- After an initial high rate of drying shrinkage concrete continues to shrink for a long period of time but at a continuously decreasing rate.

Factors affecting shrinkage of concrete:

Factor	Description
Water-Cement Ratio	Higher water-cement ratios increase shrinkage due to greater water evaporation.
Cement Content	Higher cement content generally leads to more shrinkage as hydration produces more drying.
Aggregate Content	More coarse aggregate reduces shrinkage since it acts as a restraint for the paste.
Type of Aggregate	Aggregates with lower absorption (denser) reduce shrinkage more effectively than porous aggregates.
Curing Conditions	Inadequate or improper curing leads to higher shrinkage due to rapid moisture loss.
Ambient Temperature & Humidity	High temperatures and low humidity increase evaporation rates, leading to higher shrinkage.
Mix Proportions	Concrete mixes with more fine particles (sand or silt) tend to shrink more.
Admixtures	Certain admixtures like superplasticizers can influence shrinkage behavior.
Size and Shape of the Concrete Member	Larger and more complex shapes exhibit more uneven shrinkage.
Age of Concrete	Shrinkage decreases over time, with most occurring within the first few months.
Type of Cement	Rapid-hardening cements tend to shrink more than slower-hardening varieties.
Loading Conditions	External loads can induce additional shrinkage, especially under tensile stress.